MATH O MAGIC

2·3

by

Karen Anderson

EDUCATIONAL DIRECTIONS PUBLICATIONS

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EDUCATIONAL DIRECTIONS PUBLICATIONS
11 CAVALIER COURT,
THORNLIE 6108,
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INTRODUCTION

MATH O MAGIC 2·3 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.

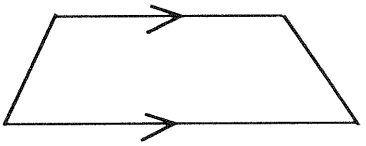
6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

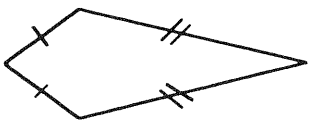
WHAT DO YOU GET IF YOU CROSS DRACULA WITH SIR LANCELOT?

Written below are some definitions. Determine the word that best fits each definition and write it in the circles. Some circles are numbered. Write the letters in the numbered circles in the code at the bottom of the page.

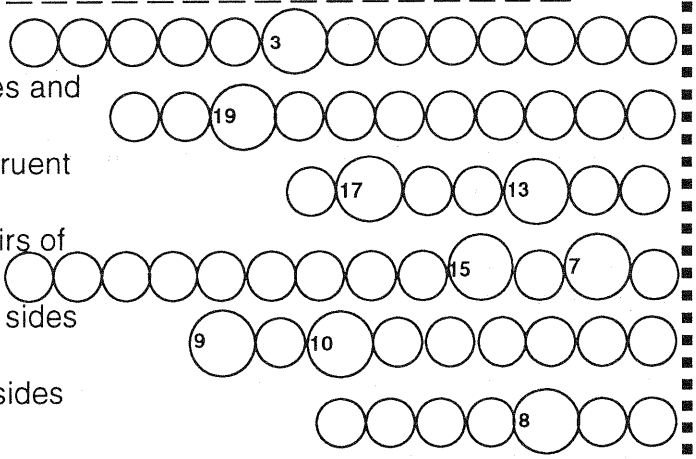
- A four sided polygon
- A triangle with congruent sides and angles
- A quadrilateral with four congruent sides
- A quadrilateral having two pairs of opposite parallel sides
- A triangle with two congruent sides
- A triangle with no congruent sides and angles is
- This quadrilateral is a



- This quadrilateral is a



- Two lines which travel in the same direction but never intersect are
- A quadrilateral with four congruent sides and four congruent angles is a
- The longest side in a right triangle is the
- Triangles which are the same shape and size are



7	13	3	18	8	16	12

2	17	9	6	20	11	15	1	14	5	10	19	4

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WHEN DRACULA WENT ON AN OCEAN CRUISE AND VISITED THE DINING ROOM, HE ASKED TO SEE . . .

14	5	9	6	15	2	10	7	11	16	3	12	4	13	8	1

Written below are some definitions. Determine the word that best fits each definition and write it in the squares. Some squares are numbered. Write the letters in the numbered squares in the code above.

- A polygon with all congruent sides and angles

	9					12

- A five sided polygon

6			1			

- A regular quatrilateral

	10					

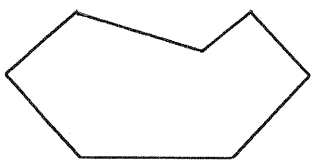
- An eight – sided polygon

						11

- A regular triangle

			13			14				

- This polygon is said to be . . .

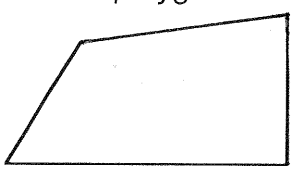


						7

- A ten – sided polygon

			15			

- This polygon is said to be . . .



				3	

- A hexagon is a polygon with

8			2				

- A seven – sided polygon can be called a septagon or a

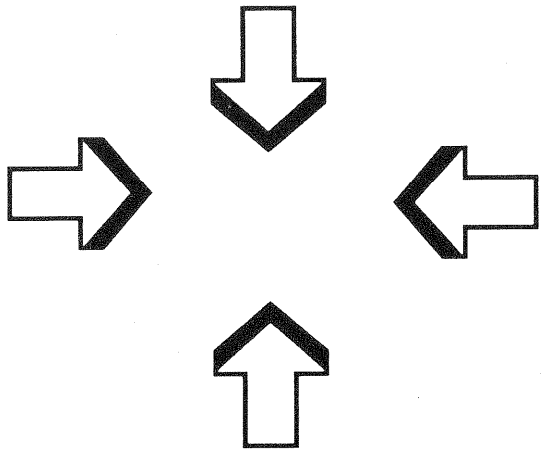
5							

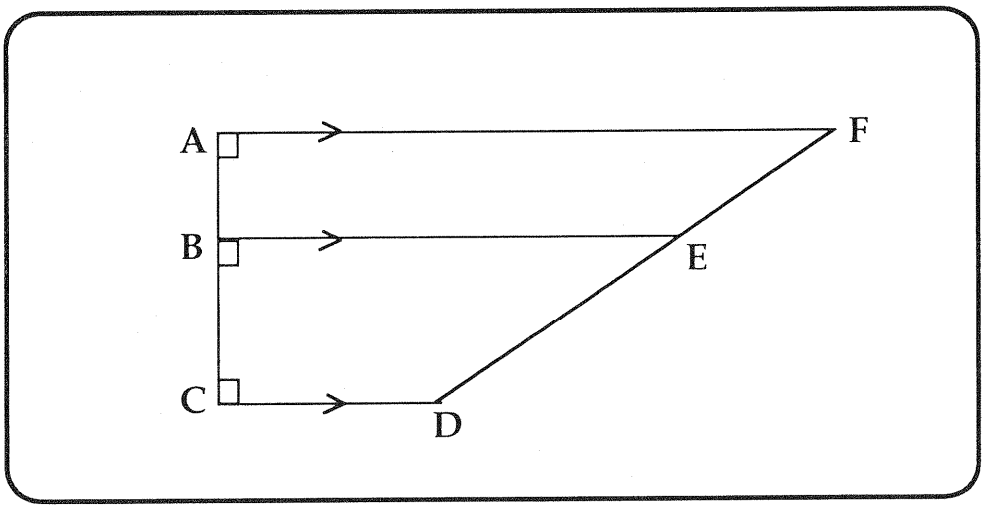
- A hexagon that has six congruent sides but not six congruent angles is

N	O	N			16		4		

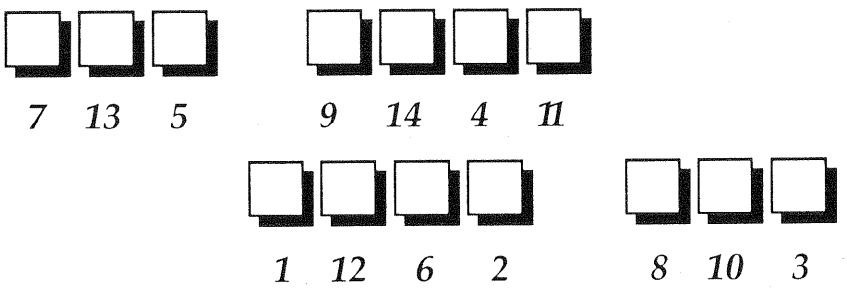
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WHICH WAY DID THE COMPUTER PROGRAMMER GO?



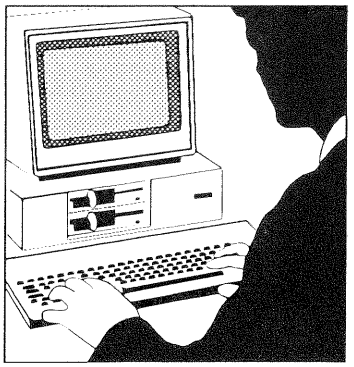


Examine the diagram above. Complete the statements on the next page by choosing the correct answer from those given. Notice the number – letter combination next to each answer. Write the letter above the number in the code at the bottom of the page.



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WHICH WAY DID THE COMPUTER PROGRAMMER GO?

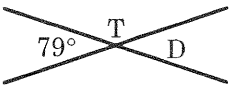


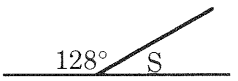
AF and CD are	parallel **10 – A**	perpendicular **2 – E**	straight **3 – M**
Angle AFE is	right **6 – F**	obtuse **4 – D**	acute **1 – D**
Angle CBE is	right **14 – E**	obtuse **3 – S**	acute **8 – R**
BE and BC are	parallel **8 – F**	perpendicular **6 – T**	straight **10 – P**
Angle CBA is	right **11 – M**	obtuse **1 – T**	straight **13 – H**
Angle BEF is	right **4 – R**	obtuse **12 – A**	acute **6 – G**
Angle ABE is congruent to	BCD **4 – N**	FEB **2 – U**	CDE **9 – T**
CD is perpendicular to	CA **3 – Y**	BE **5 – D**	DE **9 – N**
CD is parallel to	BC **7 – M**	DF **13 – S**	AF **11 – T**
Angle CDE is congruent to	FAB **14 – I**	BEF **8 – W**	BED **12 – Z**
Angle AFE is congruent to	FAB 12 – E	BEF 7 – T	BED 2 – A
FED is	right **5 – A**	obtuse **13 – R**	straight **9 – W**
BE and AF are	parallel **7 – S**	perpendicular **14 – A**	straight **11 – G**
BE and AC are	parallel **1 – F**	perpendicular **5 – E**	straight **10 – E**

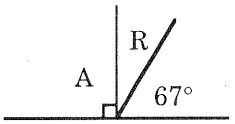
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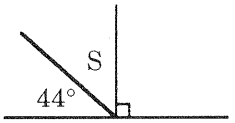
WHAT GAME DO WITCHES LIKE PLAYING?

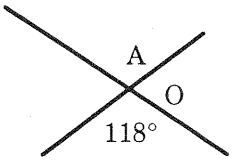


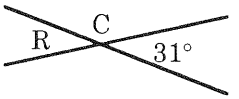


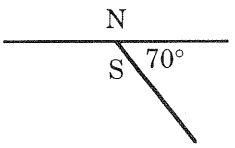


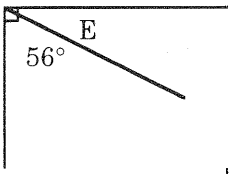


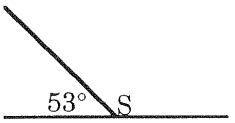


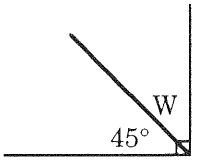




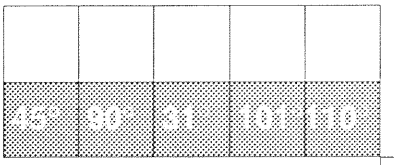




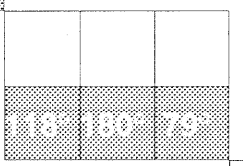




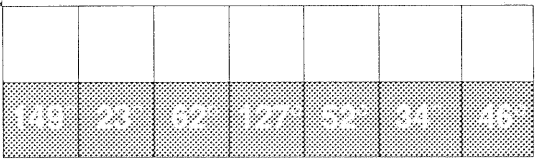
DETERMINE THE SIZES OF THE ANGLES SHOWN. LOOK FOR YOUR ANSWERS IN THE CODE BELOW. WRITE THE LETTER OF EACH ANGLE ABOVE ITS MEASURE.









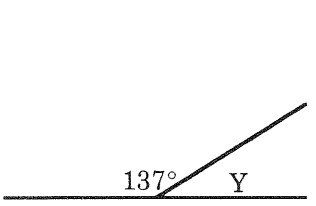


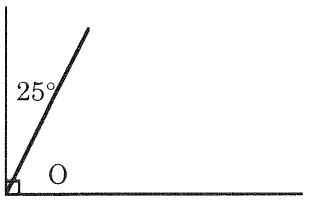
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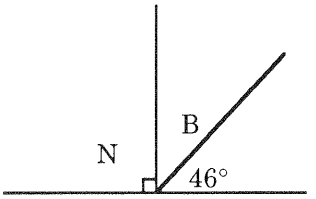
WHAT HAPPENED TO THE WOMAN WHO COVERED HERSELF WITH VANISHING CREAM?

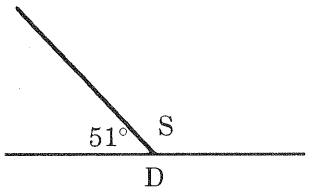
○	○	○	○	○	○	○	○	○	○	○
90°	12°	44°	74°	180°	43°	21°	79°	65°	142°	129°

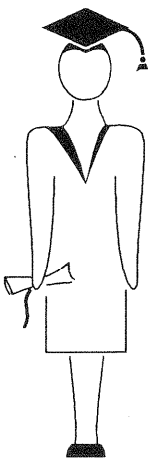
Determine the values of the unknowns in the following. Look for your answers in the code above. Write the letter of each angle above its value.











O What is the complement of 78°?

N What size angle is the complement of 11°?

W What is the supplement of 38°?

K What angle is the supplement of 159°?

O What size angle is supplementary to 106°?

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WHAT DID THE GHOST TEACHER SAY TO THE CLASS?

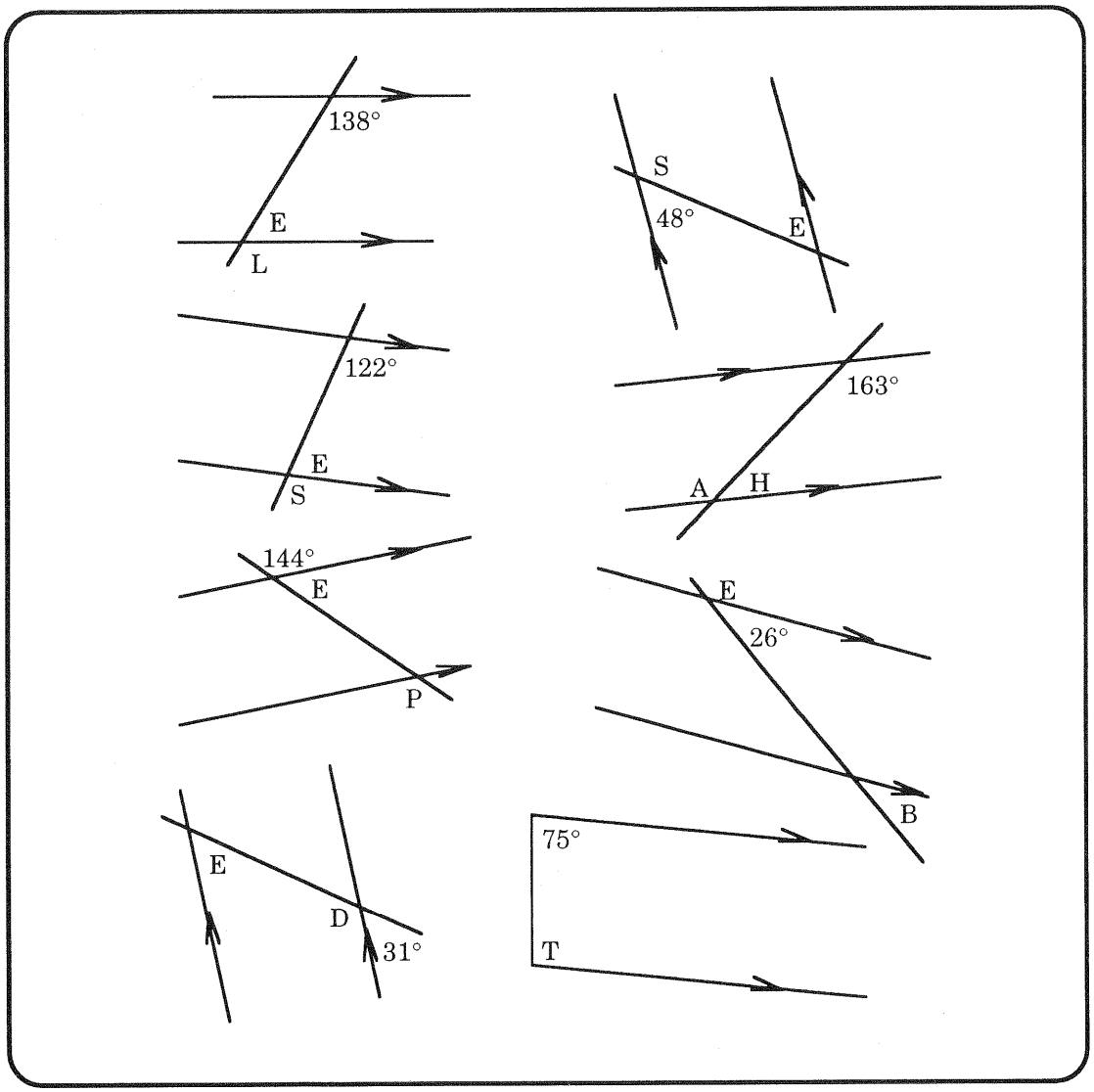


Determine the values of the unknowns below. Look for your answers in the code. Write the letter of each angle above its value.





144°	138°	31°	163°	122°	36°	26°	42°	132°	17°	48°	154°	105°	58°	149°

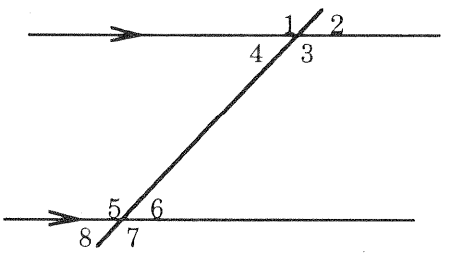


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WHAT DO YOU GET IF YOU CROSS DRACULA WITH A HAYSTACK?



USE THIS DIAGRAM TO ANSWER THE FOLLOWING QUESTIONS. LOOK FOR YOUR ANSWER IN THE CODE BELOW. WRITE THE LETTER OF EACH QUESTION ABOVE ITS ANSWER.



M If ∠4 is 22°, then ∠2 is

W If ∠4 is 25°, then ∠3 is

E If ∠6 is 48°, then ∠4 is

K If ∠8 is 71°, then ∠5 is

S If ∠8 is 55°, then ∠4 is

V If ∠7 is 116°, then ∠6 is

H If ∠7 is 163°, then ∠5 is

A If ∠3 is 151°, then ∠5 is

P If ∠1 is 126°, then∠7 is

Y If ∠2 is 33°, then ∠5 is

I If ∠2 is 59°, then ∠6 is

R If ∠3 is 144°, then ∠2 is

T If ∠1 is 127°, then ∠6 is

O If ∠8 is 67°, then ∠2 is

C If ∠1 is 99°, then ∠5 is

U If ∠7 is 170°, then ∠8 is

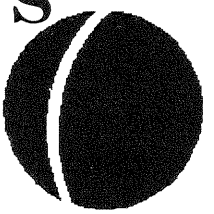
G If ∠8 is 78°, then ∠2 is

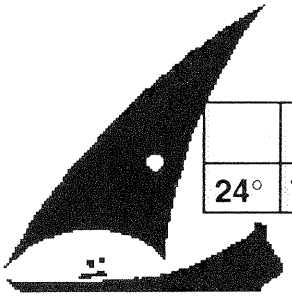


151°		64°	151°	22°	126°	59°	36°	48°		53°	163°	151°	53°
			55°	10°	99°	109°	55°		147°	67°	10°		
53°	163°	36°	67°	10°	78°	163°		151°	55°	53°	36°	151°	155°

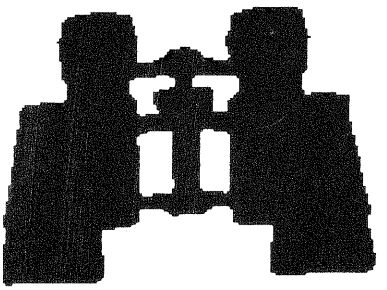
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WHAT IS A BABY GHOST'S FAVOURITE TOY?

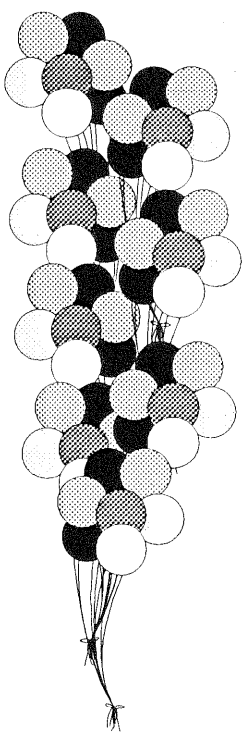


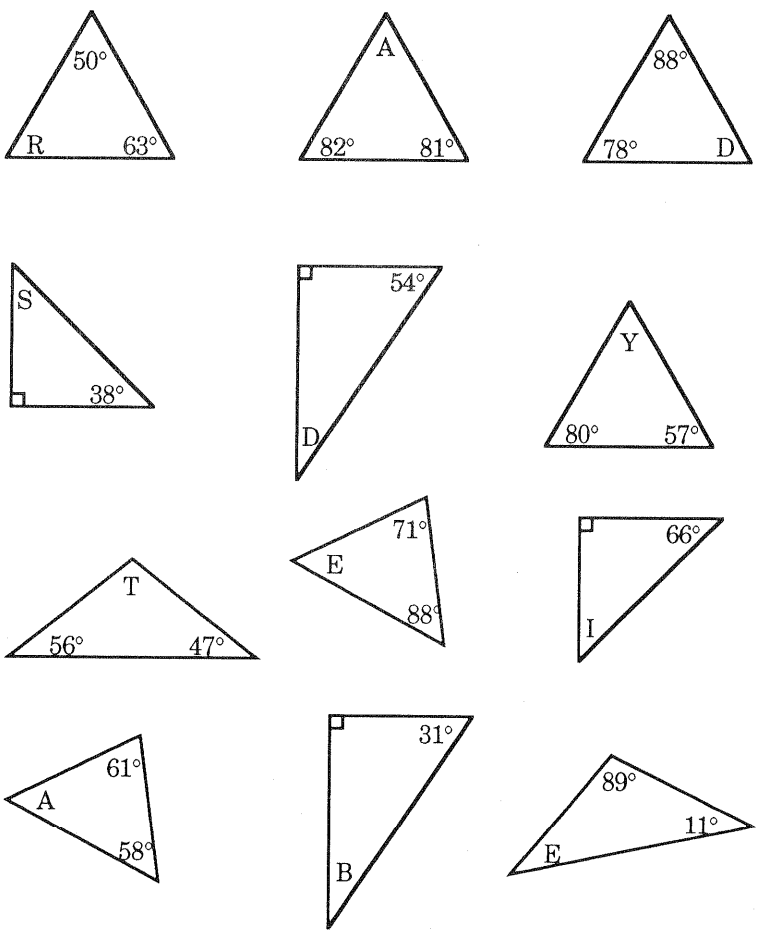


24°	77°	52°	14°	80°	17°	36°	43°	59°	21°	61°	67°



DETERMINE THE SIZES OF THE UNKNOWN ANGLES IN EACH OF THE TRIANGLES BELOW. FIND YOUR ANSWERS IN THE CODE ABOVE. WRITE THE LETTER OF EACH ANGLE ABOVE ITS SIZE.

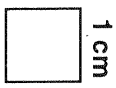




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WHICH MAGAZINE ASKED THE COMPUTER TO POSE AS ITS CENTREFOLD?

For each of the following, circle the correct answer. Notice the letter next to it. Write that letter above the number of the question in the code below.



THIS SQUARE HAS A PERIMETER OF 4 CM AND AN AREA OF 1 CM^2.

When the dimensions are doubled, the new

1. perimeter is 8 cm – **C** 16 cm – **S** 4 cm – **R**

2. area is 8 cm^2 – **E** 2 cm^2 – **T** 4 cm^2 – **A**

When the dimensions are tripled, the new

3. perimeter is 24 cm – **D** 12 cm – **P** 8 cm – **L**

4. area is 24 cm^2 – **A** 12 cm^2 – **M** 9 cm^2 – **U**

When the dimensions are quadrupled, the new

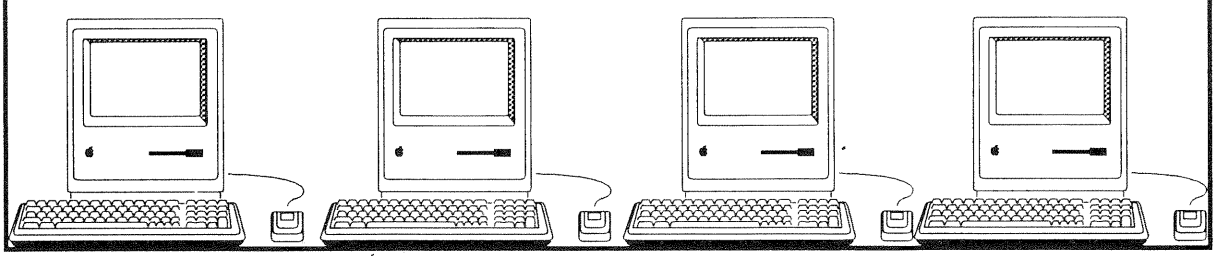
5. perimeter is 8 cm – **H** 32 cm – **L** 16 cm – **C**

6. area is 16 cm^2 – **A** 32 cm^2 – **I** 20 cm^2 – **O**

When the dimensions are halved, the new

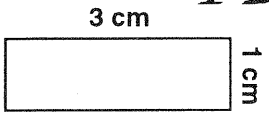
7. perimeter is 1 cm – **W** 2 cm – **S** 4 cm – **F**

8. area is 1 cm^2 – **G** 0·25 cm^2 – **E** 0·5 cm^2 – **O**



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WHICH MAGAZINE ASKED THE COMPUTER TO POSE AS ITS CENTREFOLD?



THIS RECTANGLE HAS A PERIMETER OF 8 CM AND AN AREA OF 3 CM.^{2}

When the dimensions are doubled, the new

9.	perimeter is	32 cm – **E**	8 cm – **F**	16 cm – **L**
10.	area is	9 cm^2 – **A**	12 cm^2 – **I**	6 cm^2 – **S**

When the dimensions are tripled, the new

11.	perimeter is	12 cm – **Y**	24 cm – **R**	16 cm – **B**
12.	area is	27 cm^2 – **H**	18 cm^2 – **K**	9 cm^2 – **C**

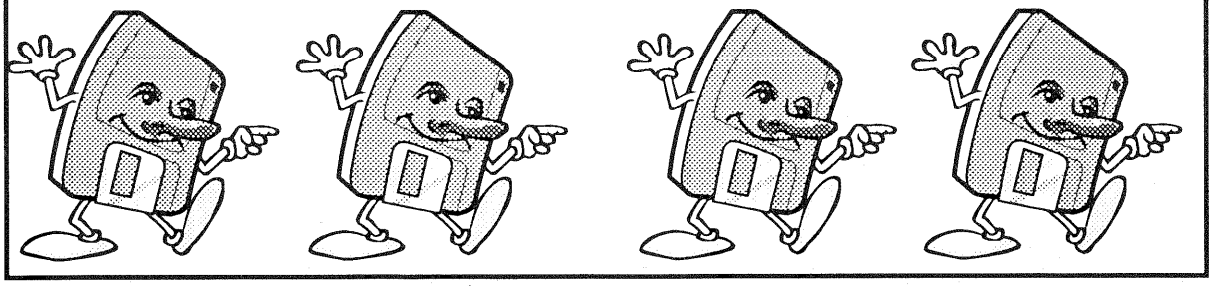
When the dimensions are quadrupled, the new

13.	perimeter is	40 cm – **U**	36 cm – **D**	32 cm – **M**
14.	area is	12 cm^2 – **T**	24 cm^2 – **L**	48 cm^2 – **P**

When the dimensions are halved, the new

15.	perimeter is	4 cm – **N**	8 cm – **D**	2 cm – **A**
16.	area is	1·5 cm^2 – **S**	0·75 cm^2 – **O**	0·5 cm^2 – **L**

3 16 14 4 9 2 11 13 8 1 12 6 15 10 5 7



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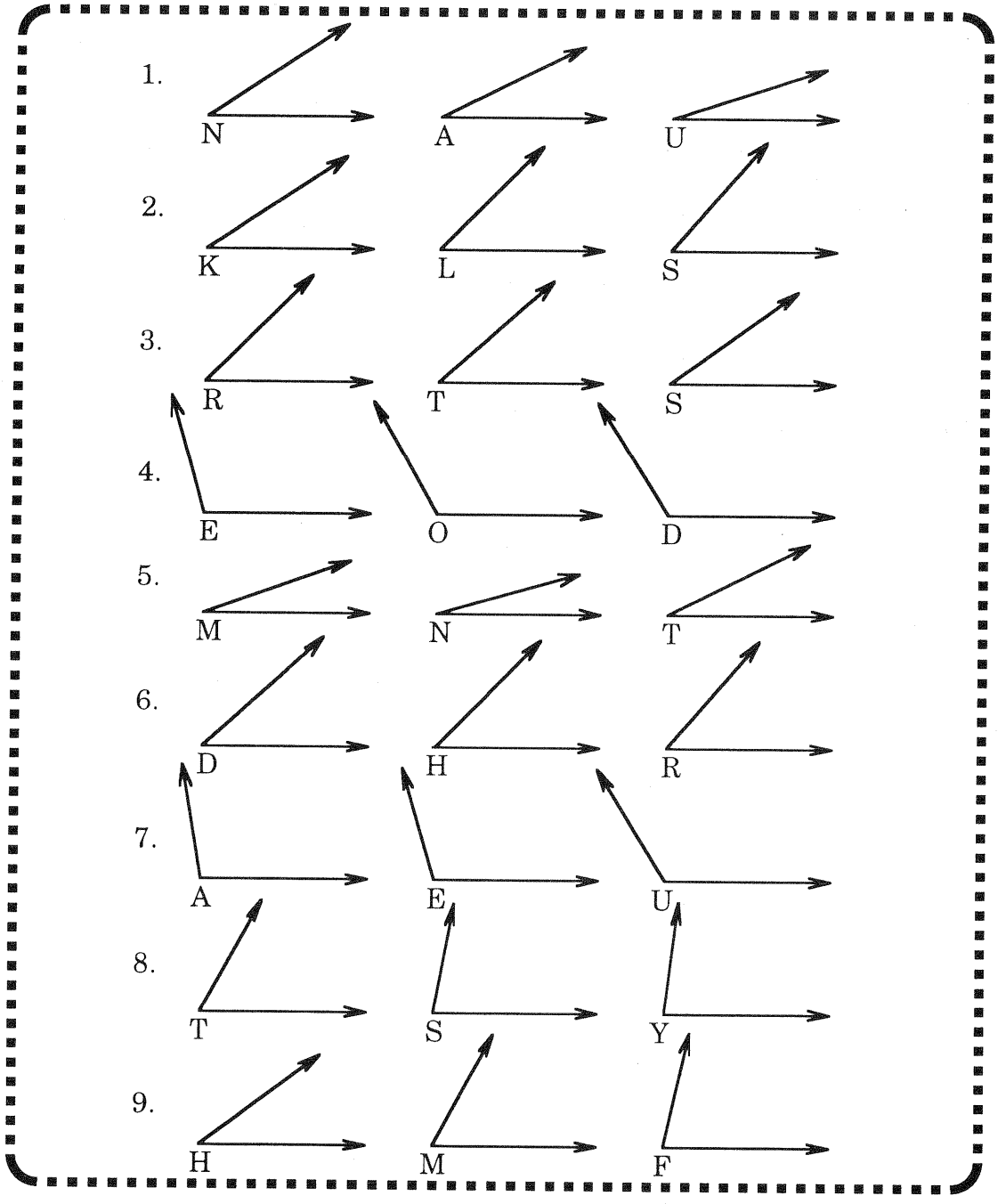


WHICH GAMES DO GHOSTS LIKE TO PLAY AT PARTIES?

Shown below are groups of 3 angles. In each group identify the smallest angle. Write the letter of that angle above the number of the question in the code.



9 7 1 5 8 7 5 6 3 4 4 2

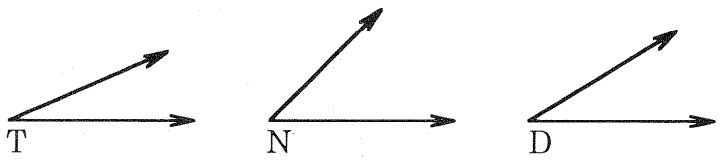


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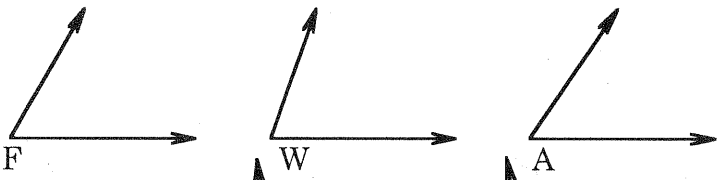
WHY DID THE BOY WEAR TWO SUITS TO THE FANCY DRESS PARTY?

Shown below are groups of 3 angles. In each group identify the largest angle. Write the letter of that angle above the number of the question in the code.

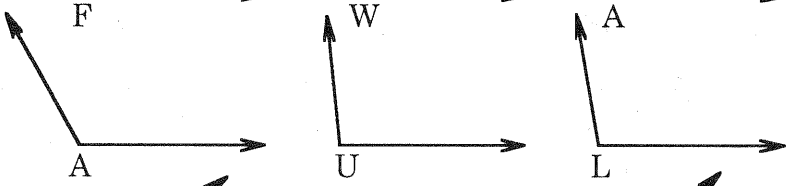
1. T N D



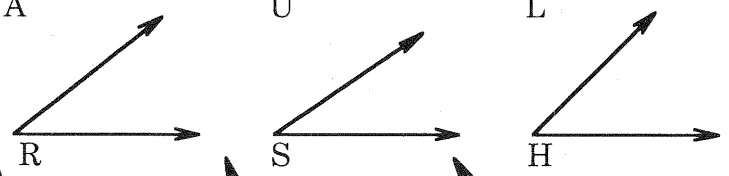
2. F W A



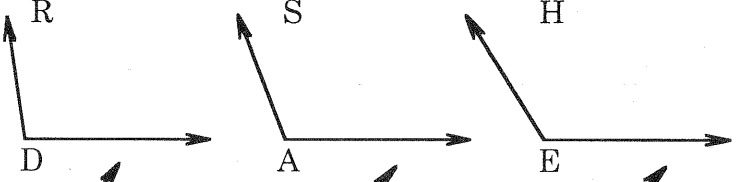
3. A U L



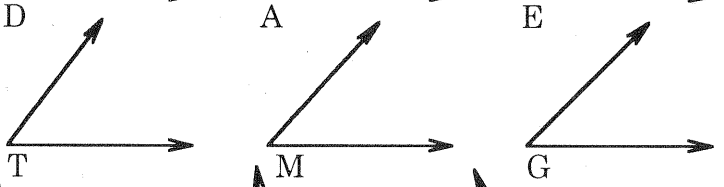
4. R S H



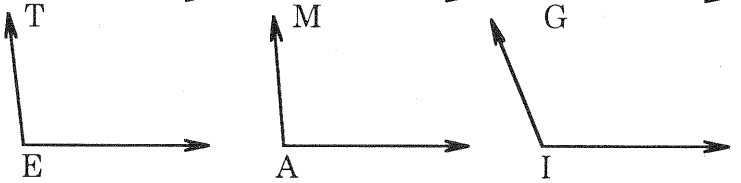
5. D A E



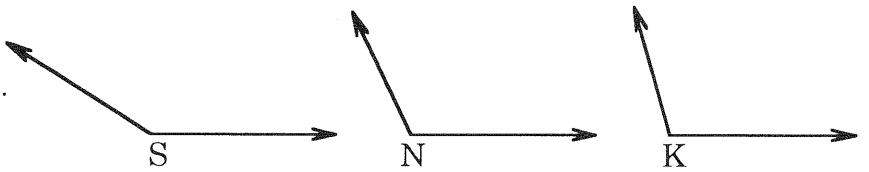
6. T M G



7. E A I



8. S N K

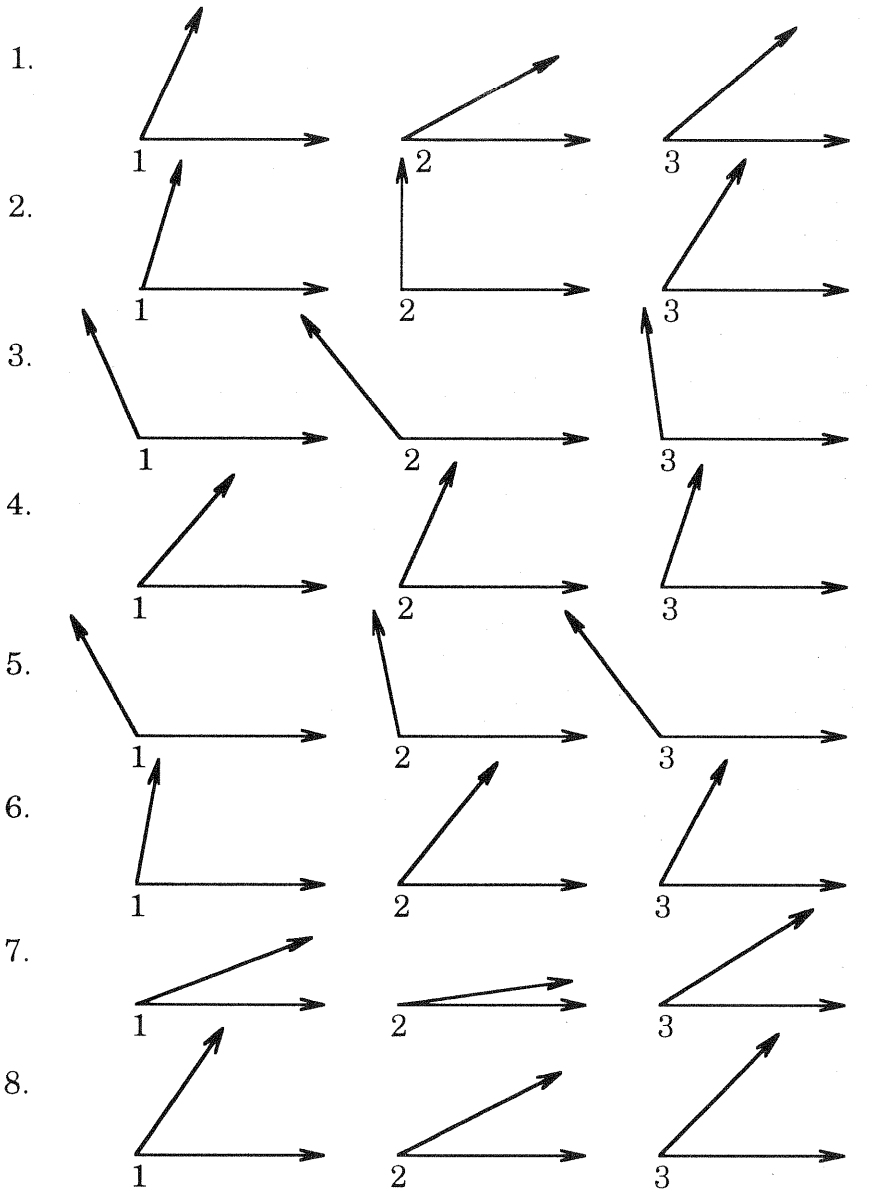


4	5	2	5	1	6	3	8	6	2	7	1	8

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WHAT IS THE DEFINITION OF A MERMAID?

Place each group of 3 angles below in order from largest to smallest. Look for the correct order in the right hand columns and circle it. Notice the letter next to each correct order. Write that letter below the number of the question in the puzzle shown below.



1.	123 – **A**	132 – **I**	321 – **E**
2.	213 – **S**	312 – **T**	231 – **W**
3.	123 – **M**	231 – **G**	213 – **H**
4.	123 – **A**	312 – **H**	321 – **E**
5.	312 – **P**	321 – **S**	213 – **P**
6.	123 – **T**	132 – **A**	213 – **O**
7.	123 – **M**	321 – **G**	312 – **F**
8.	123 – **R**	132 – **D**	213 – **B**

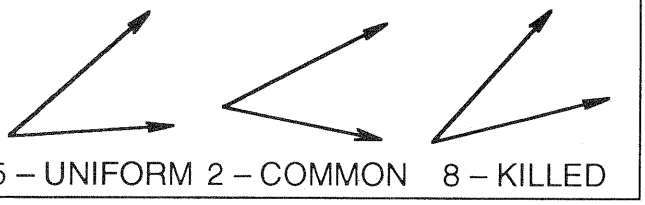
6 8 4 4 5 2 3 4 7 1 2 3

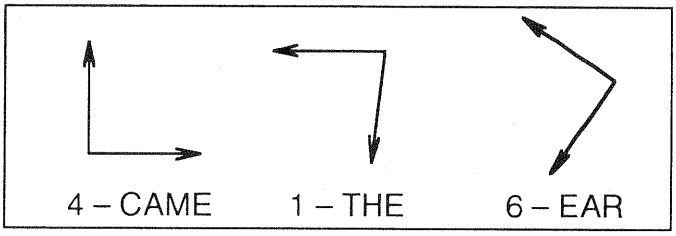


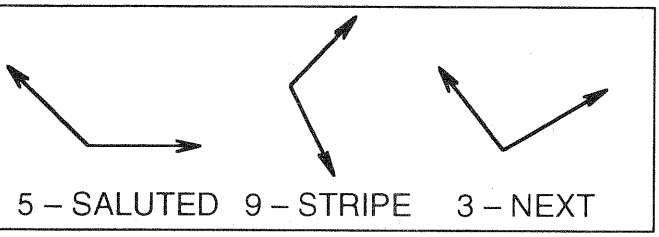
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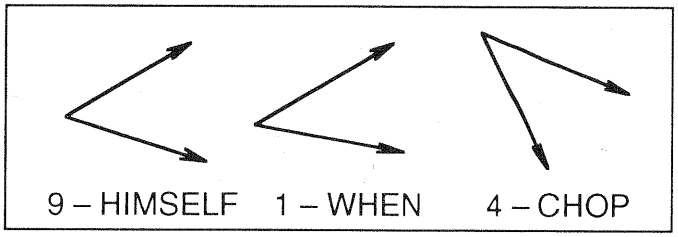
DID YOU HEAR ABOUT THE KARATE EXPERT WHO JOINED THE ARMY?

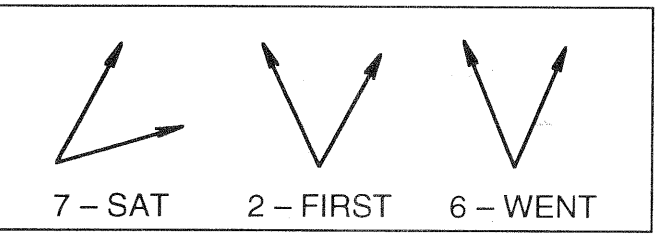
To answer this question follow these instructions. Shown below are sets of 3 angles. In each set, two of the angles are equal in size and one is different. Cross out the angles with equal sizes. Write the remaining word in the corresponding box at the bottom of the page.

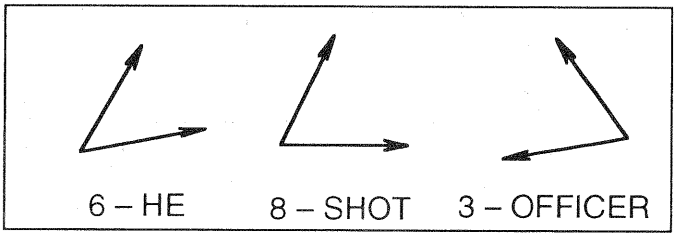


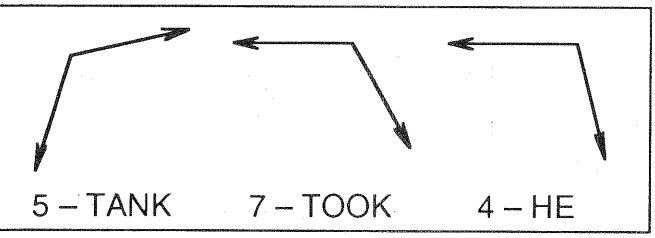


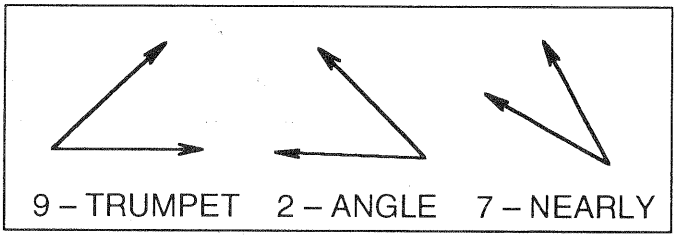




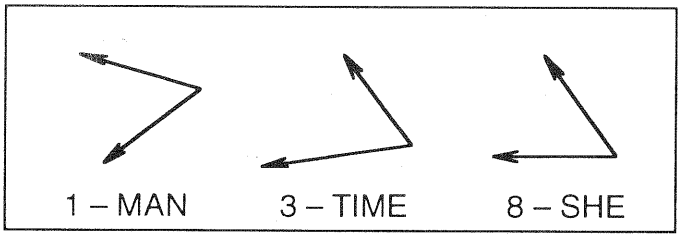












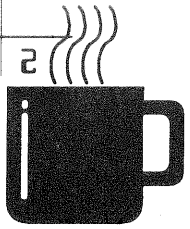


1	2	3	4	5
6	7	8	9	

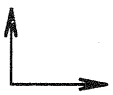
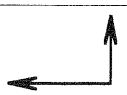
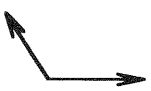
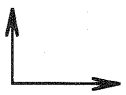
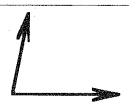
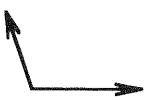
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HOW DO VISITORS FROM OUTER SPACE DRINK THEIR MILK?

7	4	12	8	7	6	3	13	1	14	2	9	5	11	10	4	2



Shown below are some angles. Determine whether each angle is acute, obtuse, right or straight. Circle the number – letter pair in the appropriate column. Write the letter above the corresponding number in the puzzle code.

	ACUTE	OBTUSE	RIGHT	STRAIGHT
	6 – **L**	2 – **E**	5 – **A**	3 – **T**
	2 – **F**	11 – **D**	4 – **R**	8 – **D**
	11 – **C**	9 – **E**	10 – **D**	13 – **E**
	1 – **A**	7 – **F**	14 – **S**	11 – **R**
	7 – **G**	3 – **O**	5 – **I**	2 – **S**
	8 – **M**	13 – **A**	6 – **N**	9 – **O**
	4 – **T**	1 – **N**	11 – **S**	12 – **M**
	9 – **U**	12 – **A**	8 – **S**	5 – **U**
	14 – **G**	4 – **S**	12 – **E**	10 – **U**
	14 – **E**	6 – **M**	9 – **A**	2 – **W**
	7 – **H**	13 – **I**	10 – **A**	14 – **H**
	5 – **S**	8 – **R**	3 – **Y**	6 – **L**
	12 – **O**	4 – **D**	7 – **A**	13 – **C**
	2 – **M**	10 – **E**	1 – **T**	3 – **D**

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HOW DO HYPNOTISTS TRAVEL?

Determine whether each of the angles shown below is acute, obtuse, right or straight. Notice the letter next to the correct answer. Write that letter above the number of the question in the puzzle below.

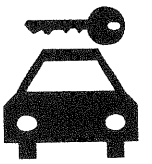
1

D – acute
P – obtuse
E – right
S – straight



2

B – acute
A – obtuse
L – right
K – straight



3

D – acute
P – obtuse
E – right
C – straight

4

S – acute
N – obtuse
T – right
W – straight

5

Y – acute
W – obtuse
E – right
D – straight



6

H – acute
S – obtuse
T – right
V – straight

7

O – acute
L – obtuse
F – right
W – straight



8

A – acute
O – obtuse
R – right
S – straight

9

Y – acute
W – obtuse
D – right
U – straight

10

K – acute
N – obtuse
L – right
S – straight

12

H – acute
S – obtuse
R – right
C – straight

11

H – acute
E – obtuse
M – right
D – straight



13

B – acute
I – obtuse
R – right
T – straight



2	5		1	9	2	10	13	3							
6	12	8	4	3	11	1	7	12	6						

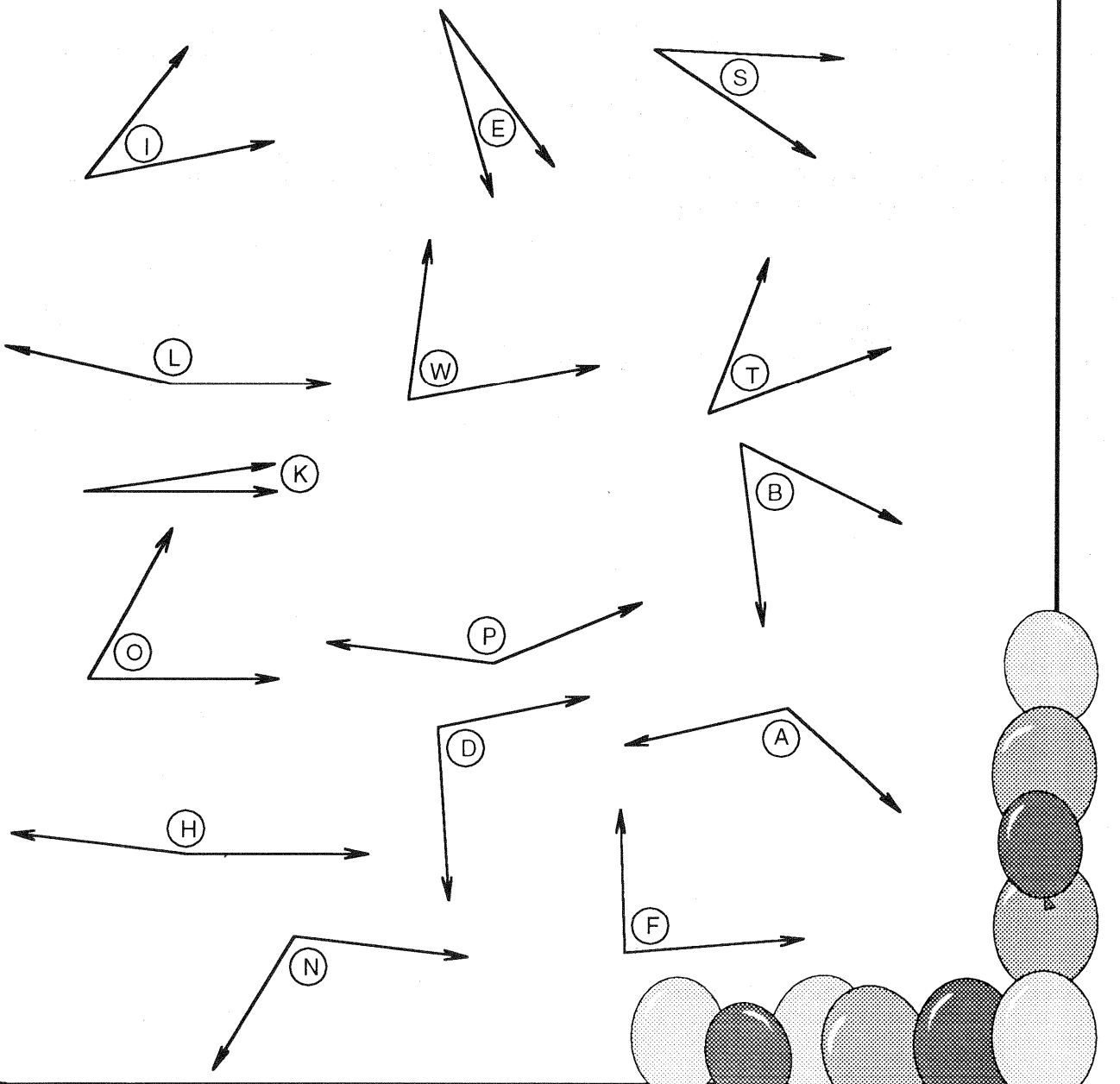
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WHAT IS THE DEFINITION OF ABSOLUTELY NOTHING?

125°	56°	125°	167°	167°	61°	61°	115°	72°	41°	49°	173°	49°	173°	20°

30°	8°	41°	115°	151°	20°	20°	167°	20°	98°	61°	87°	87°

Measure each of the following angles. Look for the size of each angle in the code above. Write the letter of each angle above its measurement.



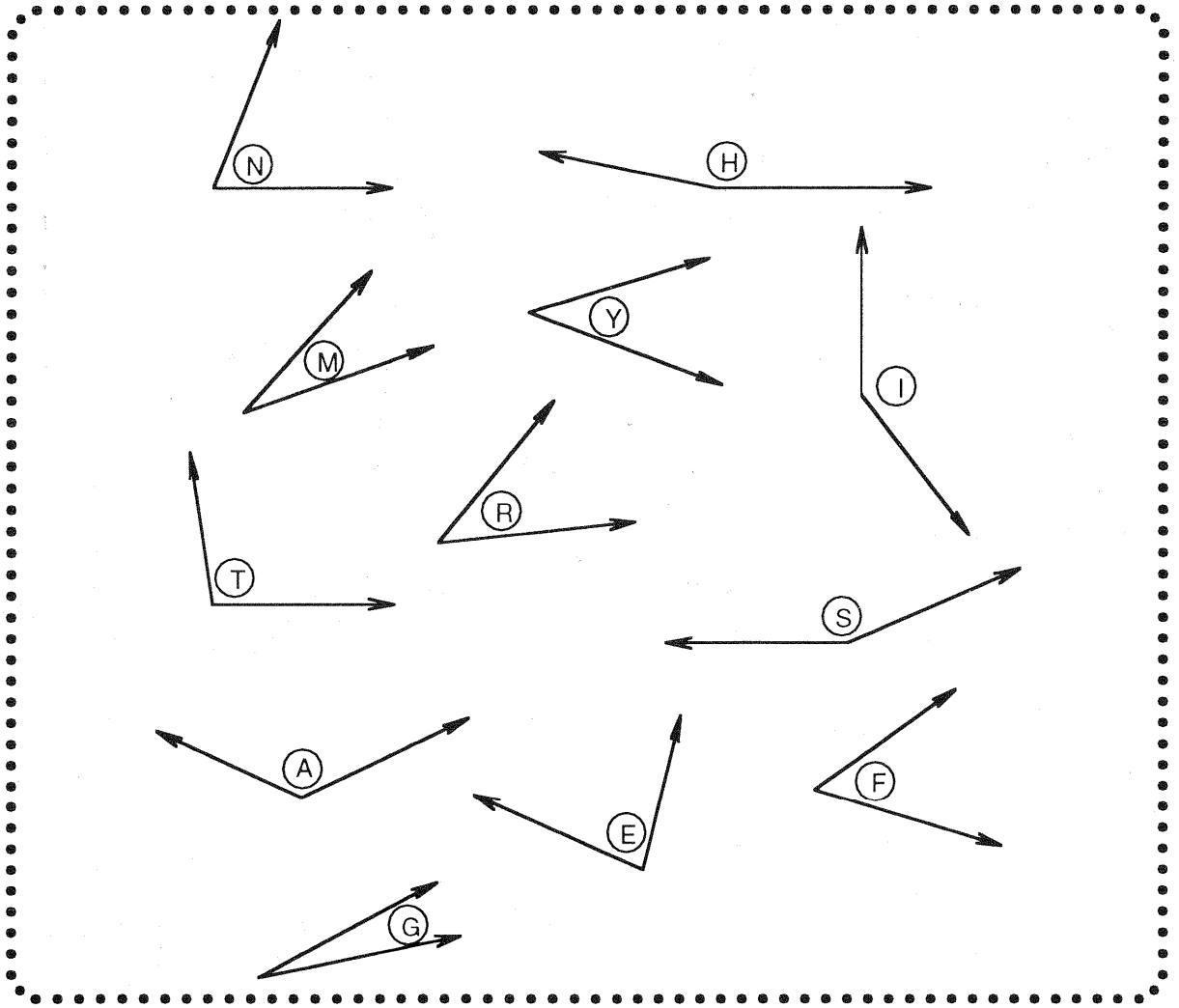
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What Did The Picture Say To The Wall?



Measure each of the following angles. Look for the size of each angle in the code . Write the letter of each angle above its measurement.

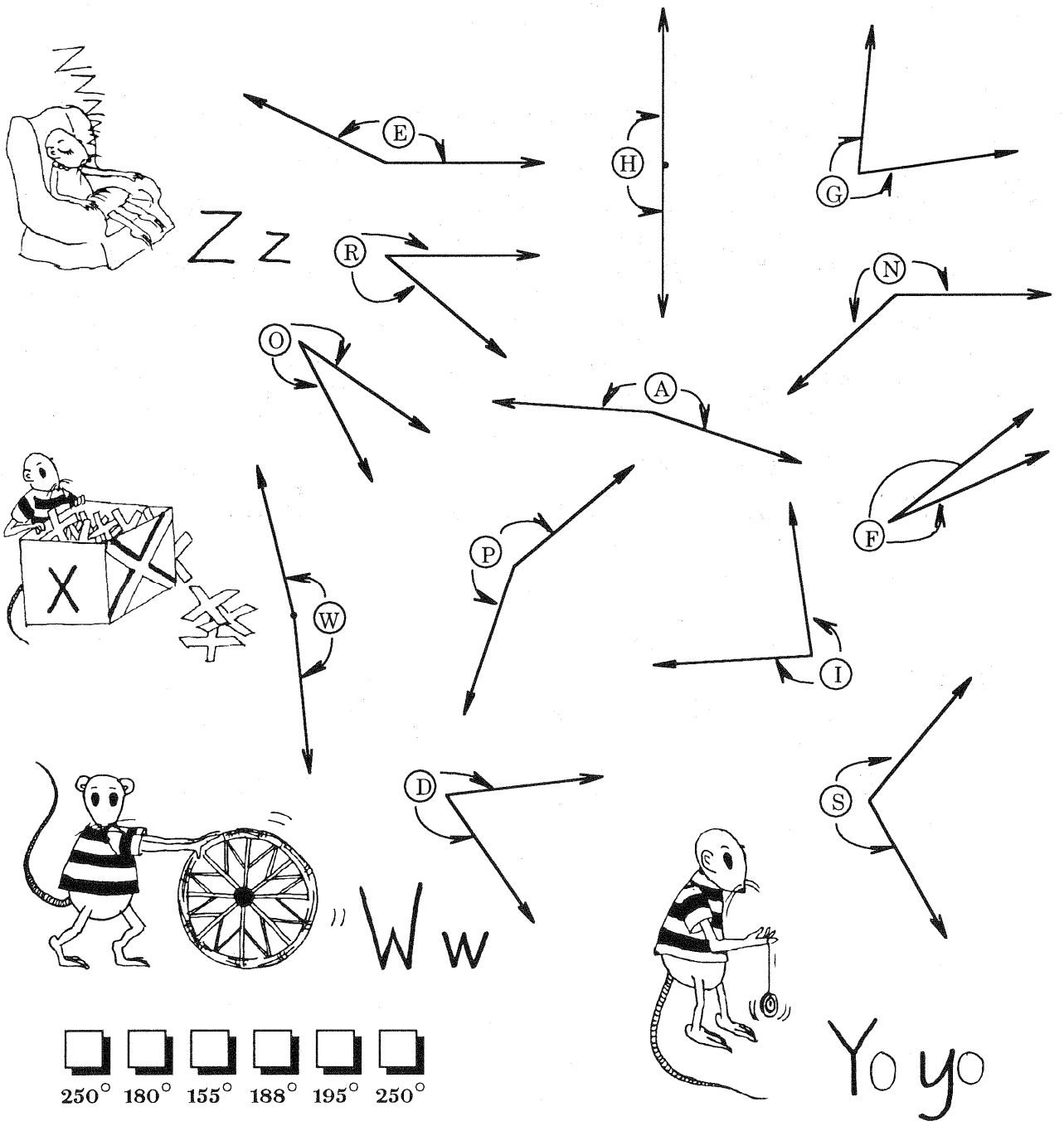
	51°	142°	45°	157°	99°	99°	169°	81°	36°	51°	45°	130°	28°	81°	
28°	81°	99°	169°	81°	68°	99°	169°	81°	36°	169°	130°	68°	17°	28°	81°



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WHY DID THE GIRL PUT HER HANDS IN THE ALPHABET SOUP?

Measure each of the following angles. Look for the size of each angle in the code below. Write the letter of each angle above its measurement.



☐	☐	☐	☐	☐	☐
250°	180°	155°	188°	195°	250°

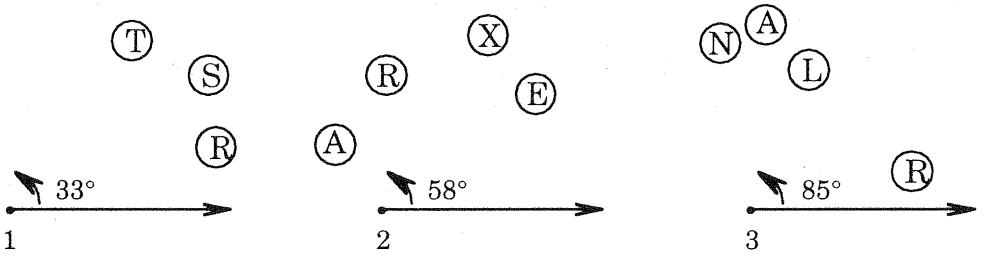
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
283°	320°	333°	210°	275°	222°	283°	345°	333°	320°	188°	333°	320°	298°	250°

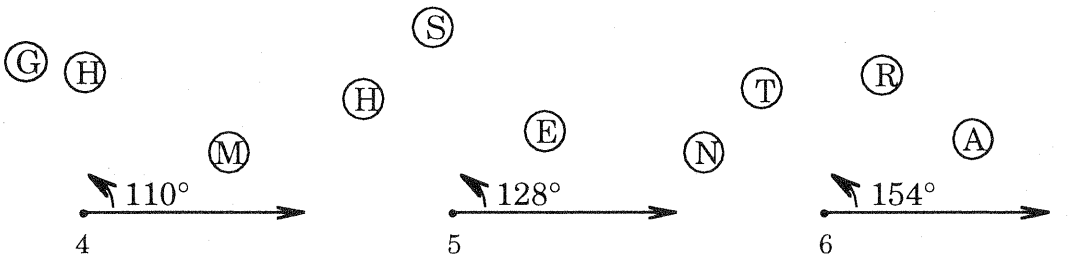
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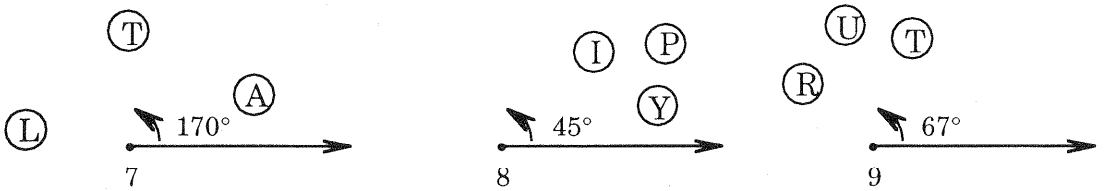
WHAT'S WORSE THAN RAINING CATS AND DOGS?



In the spaces below, one side of an angle is given. Draw in the other side. The line you draw will pass through a letter. Write the letter above the number of the angle in the puzzle code below.







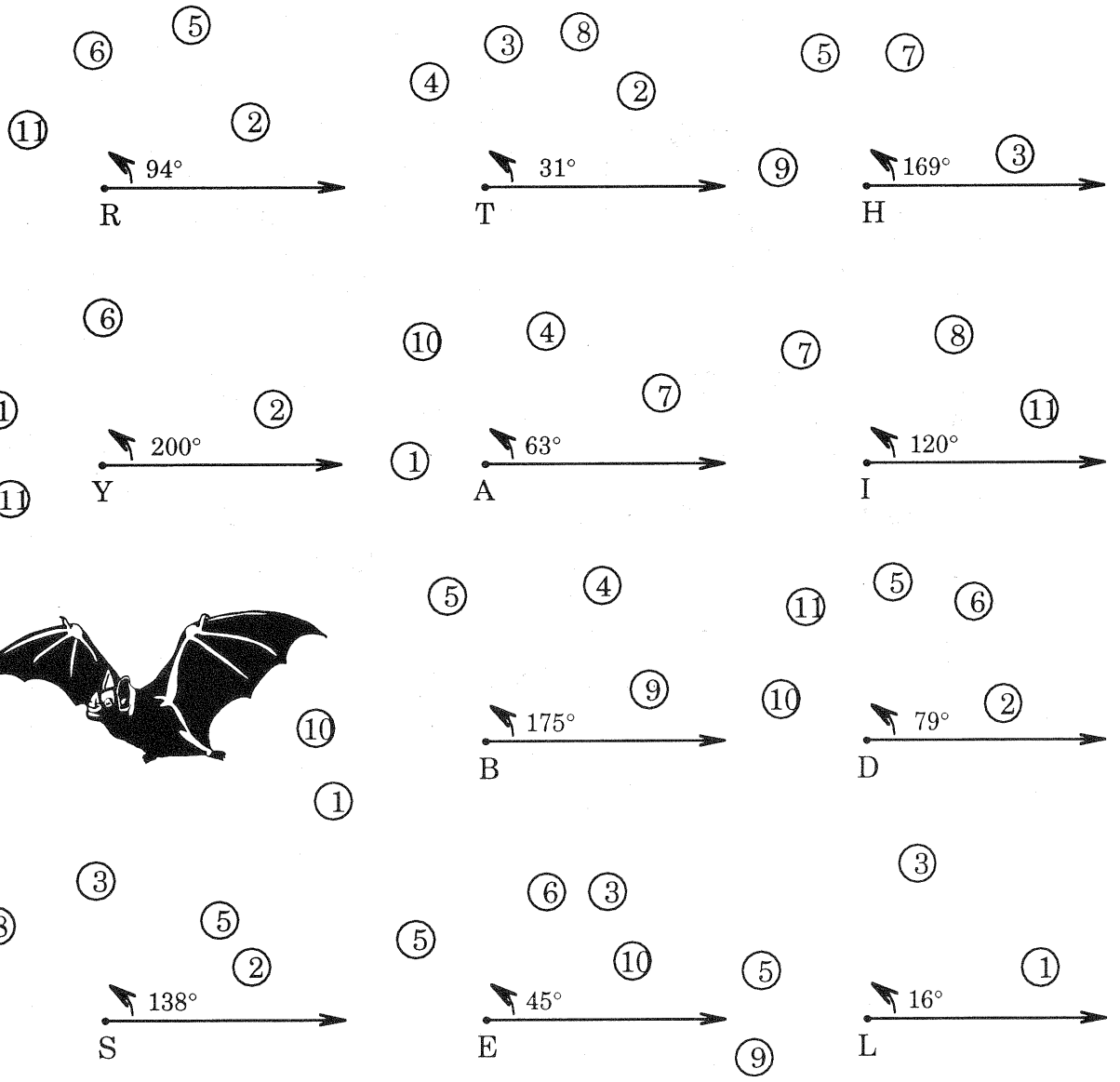
5	3	8	7	8	6	4	9	3	2	8	1	

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HOW DOES A FEMALE VAMPIRE FLIRT?

In the spaces below, one side of an angle is given. Draw in the other side. The line will pass through a number. Write the angle vertex above that number in the puzzle code below.

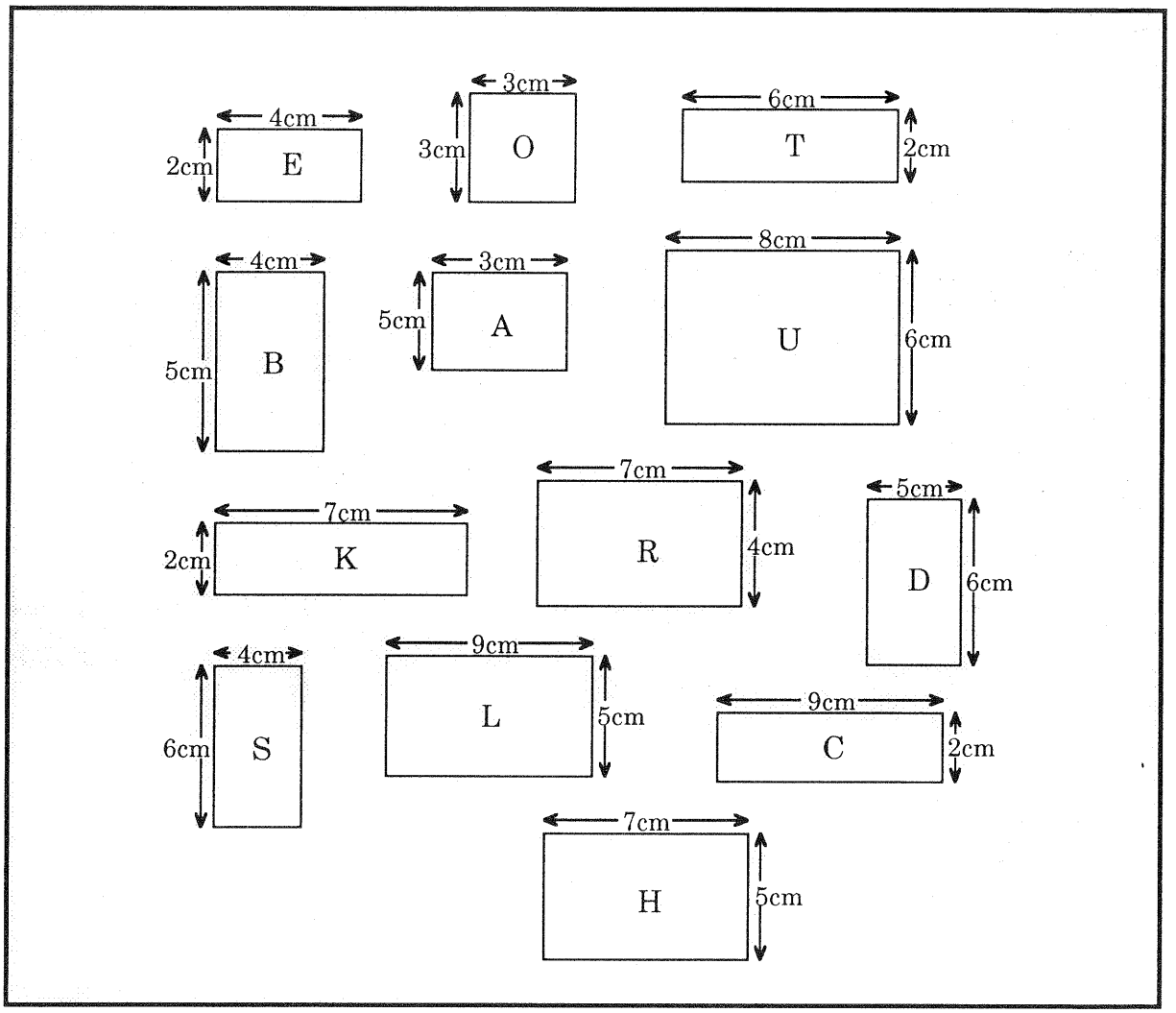


8	9	3	10	4	2	8	9	3	6	3	11	3	1	7	5	8

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Why did the prisoner want to catch measles?

DETERMINE THE AREA OF THESE RECTANGLES. LOOK FOR YOUR ANSWERS IN THE PUZZLE CODE BELOW. WRITE THE LETTER OF EACH RECTANGLE ABOVE ITS AREA.



24 cm^2	9 cm^2		35 cm^2	8 cm^2		18 cm^2	9 cm^2	48 cm^2	45 cm^2	30 cm^2
20 cm^2	28 cm^2	8 cm^2	15 cm^2	14 cm^2		9 cm^2	48 cm^2	12 cm^2		

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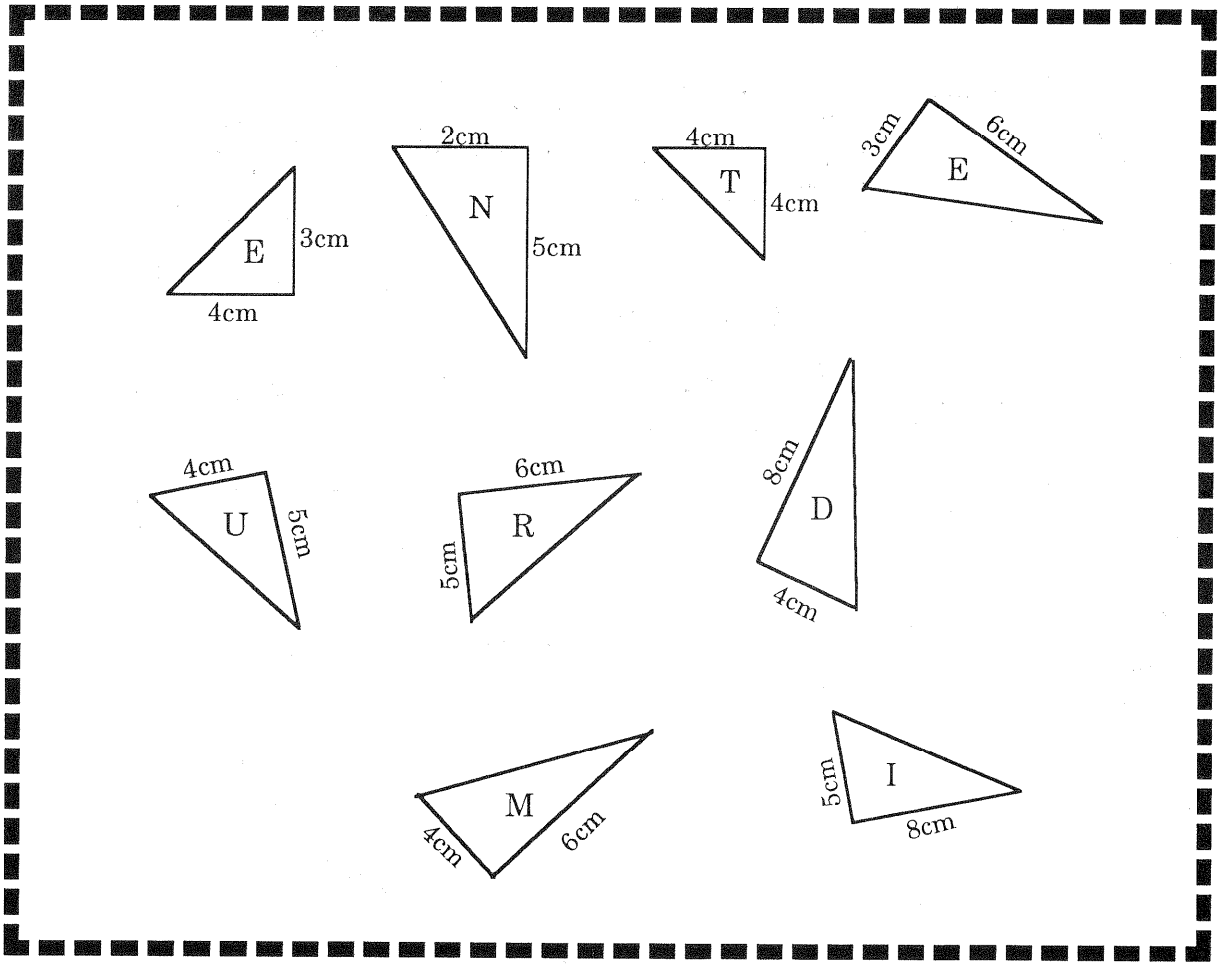


WHAT DOES A DIVER GET PAID WHEN HE WORKS EXTRA HOURS?

$10\ cm^2$	$5\ cm^2$	$16\ cm^2$	$9\ cm^2$	$15\ cm^2$	$8\ cm^2$	$20\ cm^2$	$12\ cm^2$	$6\ cm^2$

Determine the area of the following triangles. Look for your answers in the puzzle code above. Write the letter of each triangle above its area.





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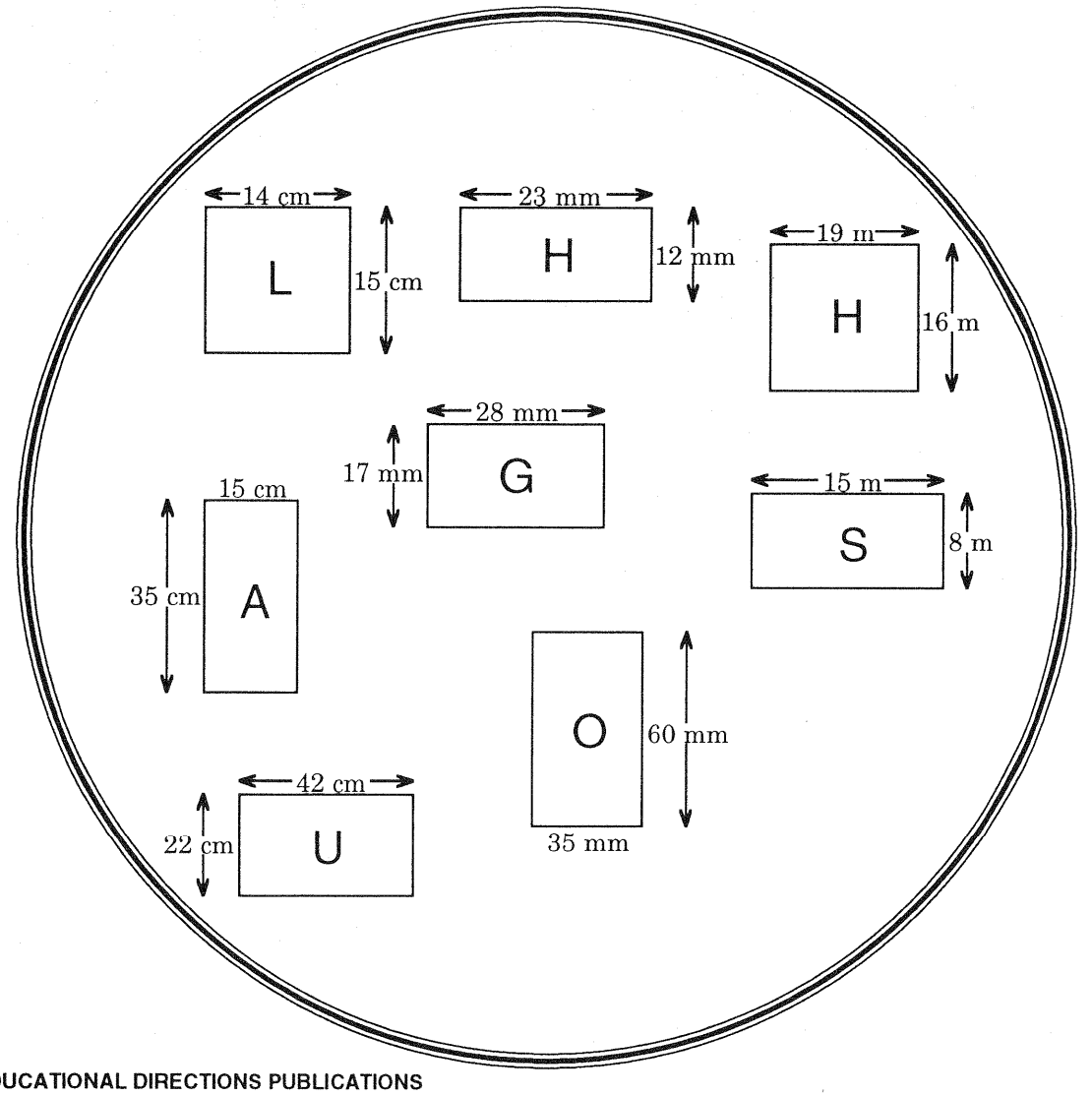
WHAT IS A GHOST'S FAVOURITE FOOD?

FIND THE AREA OF THE FOLLOWING RECTANGLES.

LOOK FOR YOUR ANSWERS IN THE CODE BELOW.

WRITE THE LETTER OF EACH RECTANGLE ABOVE ITS AREA.

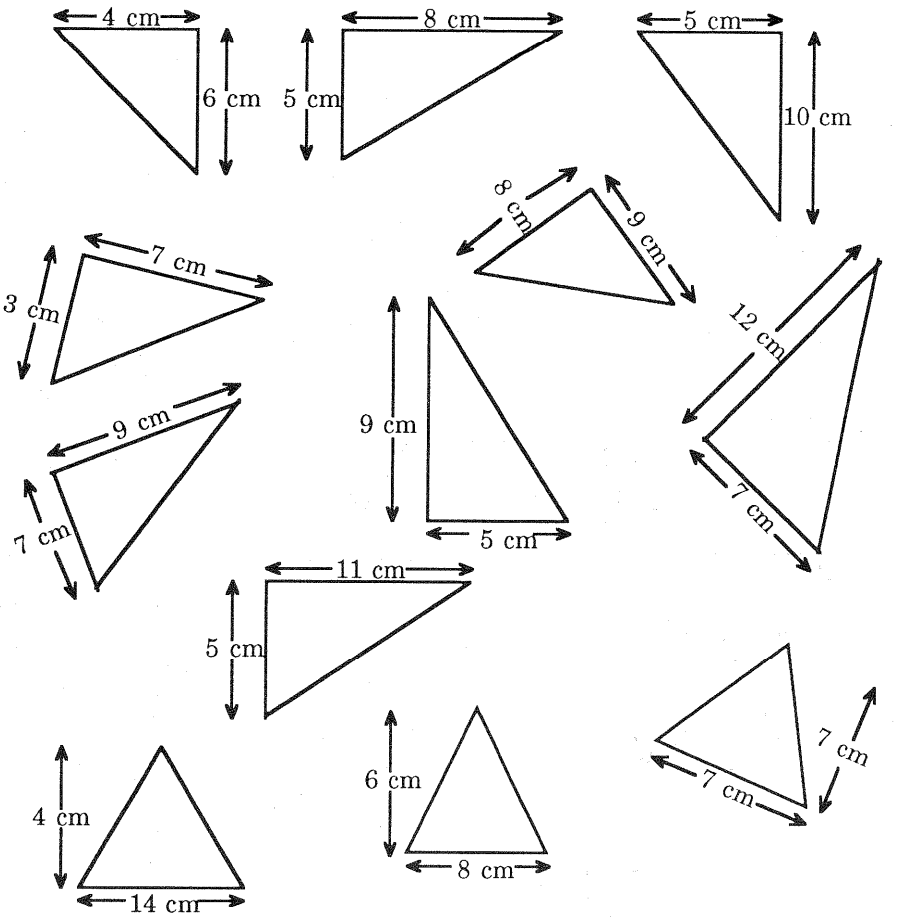
476 mm^2	304 m^2	2100 mm^2	924 cm^2	210 cm^2	525 cm^2	120 m^2	276 mm^2



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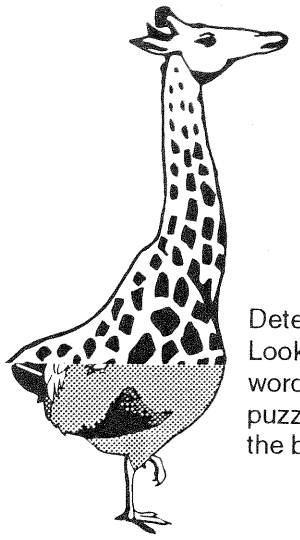
WHAT DO YOU GET WHEN YOU CROSS AN OCTOPUS WITH A COW?

Determine the area of the following triangles. Look for your answers in the grid below. Cross out each box containing a correct answer. When you finish there will be 6 boxes remaining. Read the words in those boxes in order from left to right.



THE	EIGHT	AN	BULL	ANIMAL	THEN
10·5 cm^2	31·5 cm^2	44 cm^2	12 cm^2	18 cm^2	22·5 cm^2
THAT	**UDDER**	**ARMS**	**AND**	**CAN**	**WATER**
8 cm^2	42 cm^2	25 cm^2	28 cm^2	40 cm^2	24 cm^2
CREAM	**MILK**	**TO**	**ITSELF**	**LEGS**	**CARTON**
20 cm^2	26 cm^2	27·5 cm^2	45 cm^2	24·5 cm^2	36 cm^2

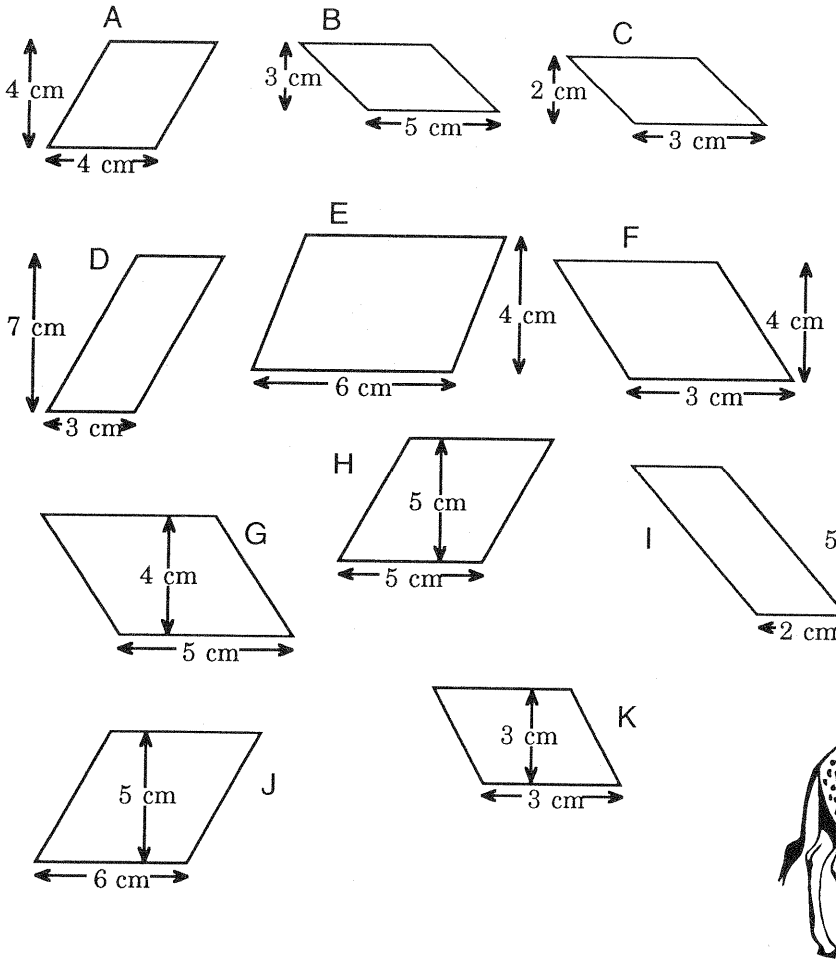
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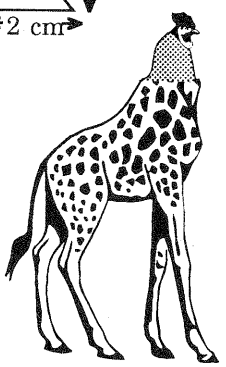


WHAT WOULD YOU GET IF YOU CROSSED A GIRAFFE WITH A ROOSTER?

Determine the area of the following parallelograms. Look for your answers in the Code Key. Notice the word next to each answer. Write that word in the puzzle box containing the letter of the question at the bottom of the page.

CODE KEY	
24 cm^2 –	PEOPLE
22 cm^2 –	NECK
16 cm^2 –	AN
25 cm^2 –	ON
15 cm^2 –	ANIMAL
18 cm^2 –	CROW
9 cm^2 –	FLOOR
6 cm^2 –	THAT
20 cm^2 –	LIVE
26 cm^2 –	DAYBREAK
30 cm^2 –	TOP
21 cm^2 –	WAKES
32 cm^2 –	SLEEP
36 cm^2 –	BED
12 cm^2 –	WHO
10 cm^2 –	THE
5 cm^2 –	LONG



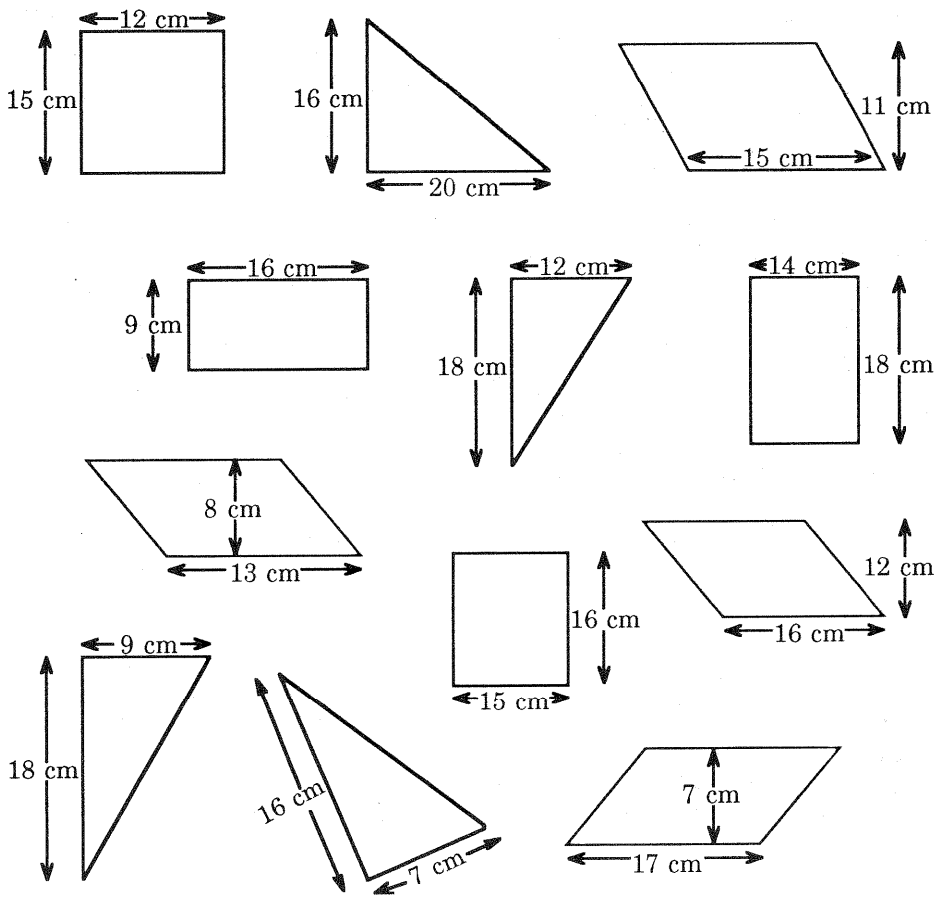


A	B	C	D	E	F
G	H	I	J	K	

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WHAT DO YOU GET IF YOU CROSS A HOMING PIGEON WITH A PARROT?

Determine the area of each of the following. Look for your answers in the grid shown below. Cross out each box containing a correct answer. When you finish there will be 12 boxes remaining. Read the words in those boxes in order from left to right.



AN	A	HOW	BIRD	SEEDS	FLY
240 cm^2	92 cm^2	104 cm^2	162 cm^2	81 cm^2	252 cm^2
THAT	SKY	CAN	ASK	TALK	FEATHERS
216 cm^2	192 cm^2	98 cm^2	224 cm^2	144 cm^2	165 cm^2
SQUAWK	ITS	THEN	WAY	WILL	HOME
180 cm^2	250 cm^2	160 cm^2	105 cm^2	119 cm^2	112 cm^2
IF	IT	TO	GETS	AIR	LOST
320 cm^2	175 cm^2	56 cm^2	220 cm^2	108 cm^2	190 cm^2

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WHERE DO BLACKBIRDS DRINK?

Determine the area of each of the shapes overleaf. Look for your answers in the puzzle grid. Cross out each box containing a correct answer. When you finish there will be 5 boxes remaining. Write the letters in those boxes in order from left to right.

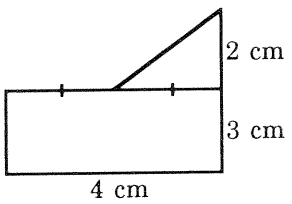


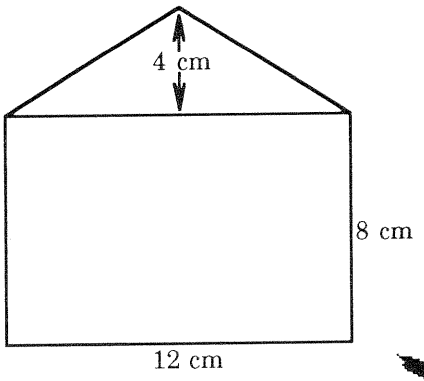
TH 40 cm^2	AT 90 cm^2	EN 14 cm^2	AC 38 cm^2
ST 80 cm^2	EA 34 cm^2	RO 110 cm^2	RT 120 cm^2
WB 60 cm^2	AM 65 cm^2	AR 30 cm^2	MB 27 cm^2

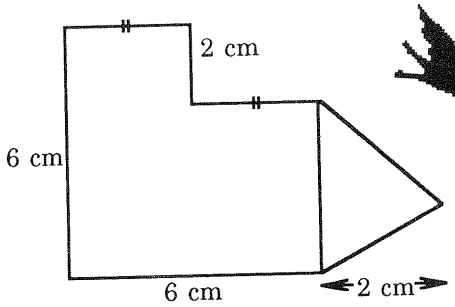
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WHERE DO BLACKBIRDS DRINK?

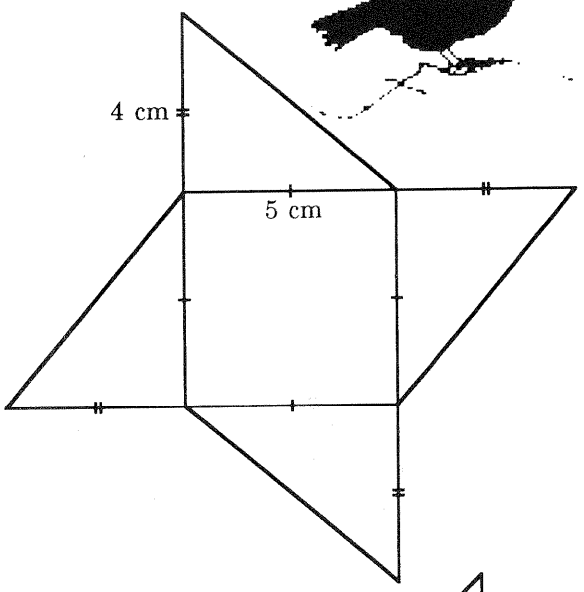


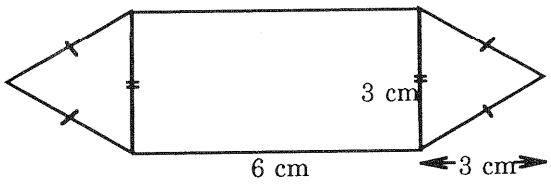


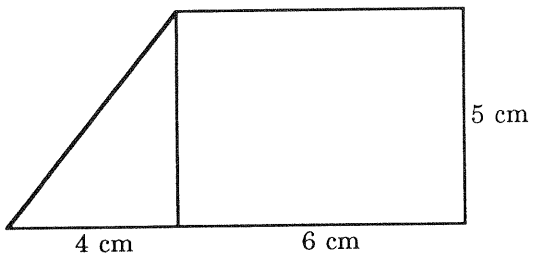


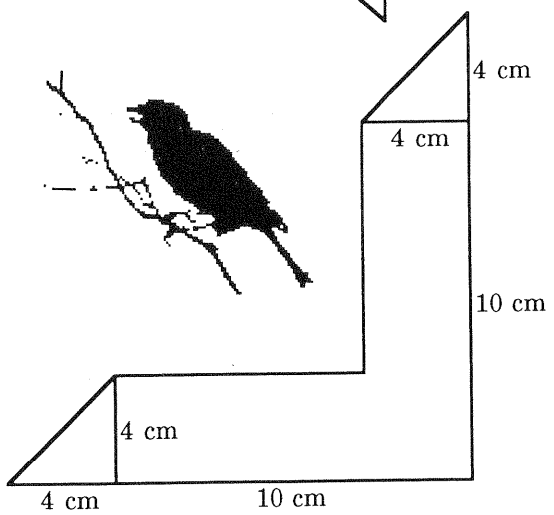








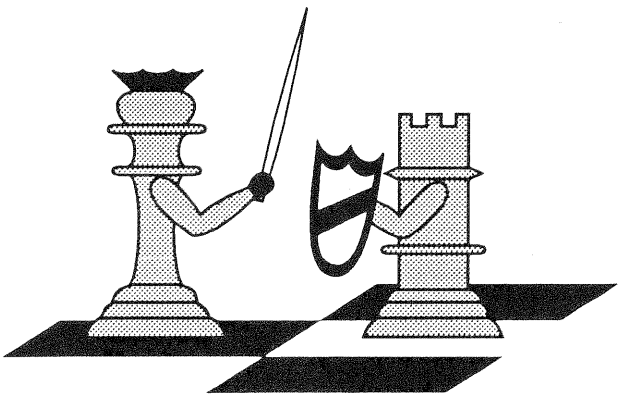






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WHAT WEAPON DO KNIGHTS FEAR THE MOST?

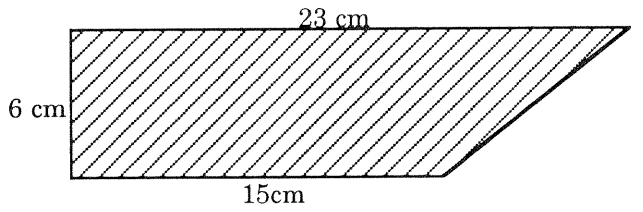


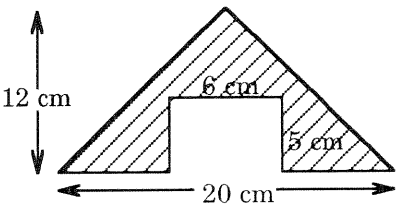
Find the area of the shaded regions shown on the next page. Look for your answers in the puzzle grid. Cross out each box containing a correct answer. When you finish there will be 5 boxes remaining. Write the letters in those boxes in order from left to right.

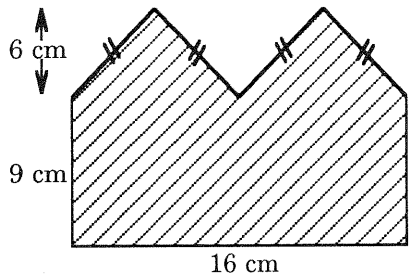
64·5 cm^2 AC	107 cm^2 ST	90 cm^2 RE	125 cm^2 AN
82·5 cm^2 LU	98·5 cm^2 OP	114 cm^2 CK	52 cm^2 RL
120 cm^2 EN	102 cm^2 SA	105 cm^2 ER	192 cm^2 EM

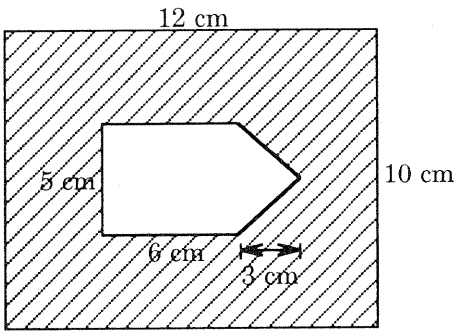
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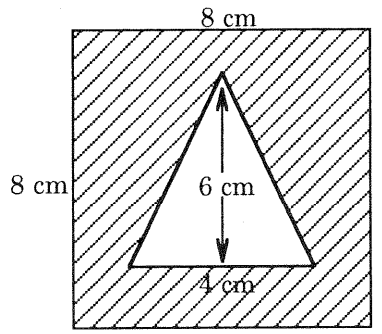
WHAT WEAPON DO KNIGHTS FEAR THE MOST?

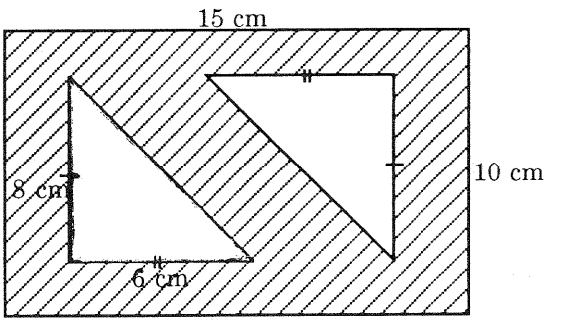


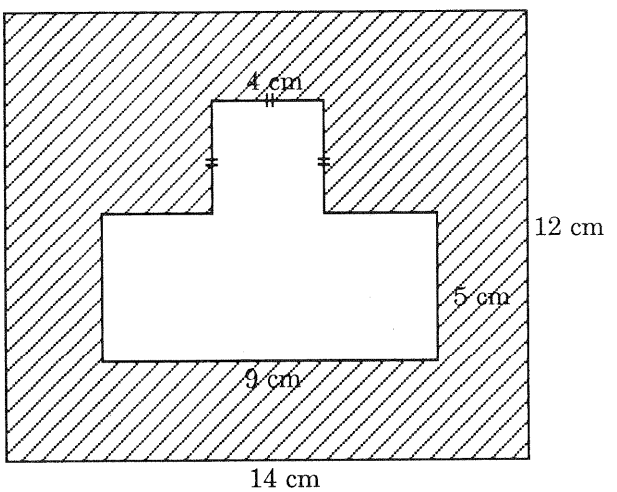


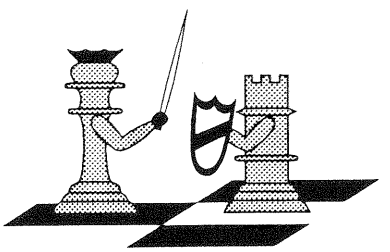










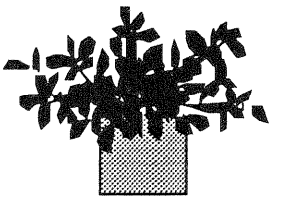


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WHAT IS AUSTRALIA'S MOST POPULAR GARDENING MAGAZINE?



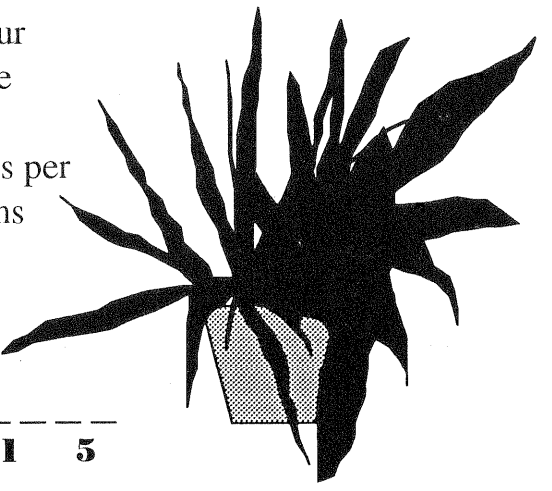
Solve each of the following. Look for your answer in the CODE KEY. Notice the letter next to it. Write that letter above the number of the question in the puzzle below.



1. Dean types at a rate of 42 words per minute. How many words could he type in 10 minutes?
2. Kerry's pulse was measured as 19 beats in 15 seconds. What is her pulse rate per minute?
3. Steak was being sold at $7·80 per kilogram. How much would 3 kilograms cost?
4. At a speed of 70 km/h, how far would a car travel in 5 hours?
5. Water was filling a 9 litre bucket every 30 seconds. How long would it take to fill a 45 litre container?
6. A 250 g box of chocolates costs $5·25. How much would 1 kg of the same chocolates cost?
7. At a rate of 76·5 cents/litre of petrol, how much would it cost to fill a 20 litre container?
8. A patient was required to take 5 mL of medicine every 2 hours. Over a 24 hour period, how much medicine would she have taken?
9. A cricket team's run rate was 4·36 runs per over. In a 50 over game how many runs did it score?

CODE KEY

60 mL	I
76	W
95	B
350 km	H
420	R
$21	E
109	K
$2·60	N
$23·40	G
5 minutes	Y
218	D
2·5 minutes	S
$15·30	T



7	4	6	2	6	6	9	6	1	5

9	8	3	6	5	7

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WHY WAS THE HUMAN CANNONBALL NO LONGER WORKING AT THE CIRCUS?

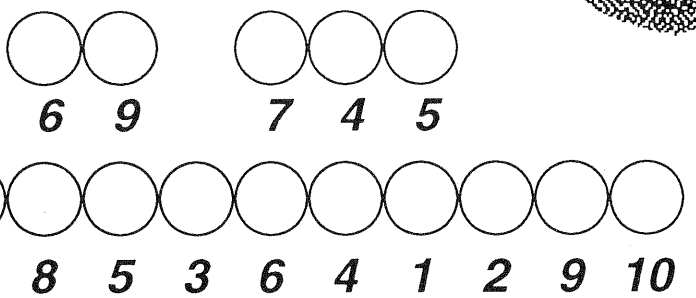
Calculate the mean of each of the following. Look for your answer in the code below. Write the letter next to each answer above the number of the question.

1. 12, 15, 16, 16, 26
2. 7, 16, 20, 25, 28, 30
3. 6, 10, 11, 11
4. 2, 8, 16, 17, 17, 20, 22, 22
5. 10, 12, 15, 25, 30, 41, 63
6. 12·5, 16·5, 18·5, 19·5, 31, 40·5, 42·5, 43, 46
7. 66, 48, 52, 51, 43
8. 99, 86, 79
9. 88, 190, 150, 102, 97, 117
10. 122, 138, 146, 141, 169, 113, 158



CODE KEY

28	– **S**	30	– **H**
141	– **D**	88	– **I**
21	– **G**	15·5	– **A**
17	– **R**	124	– **E**
52	– **W**	9·5	– **C**



6 9 7 4 5

10 8 5 3 6 4 1 2 9 10

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HOW DOES A VAMPIRE BEGIN ITS LETTERS?

DETERMINE THE MEDIAN OF EACH OF THE FOLLOWING. LOOK FOR YOUR ANSWERS IN THE CODE KEY. NOTICE THE LETTER NEXT TO EACH ANSWER. WRITE THAT LETTER ABOVE THE NUMBER OF THE QUESTION IN THE PUZZLE CODE ON THE NEXT PAGE.

1. 3, 4, 6, 8, 9

2. 28, 35, 36, 37, 38, 42, 45

3. 45, 47, 48, 49, 50, 55

4. 18, 28, 23, 11, 13

5. 58, 58, 48, 60, 52, 51, 57, 52, 59



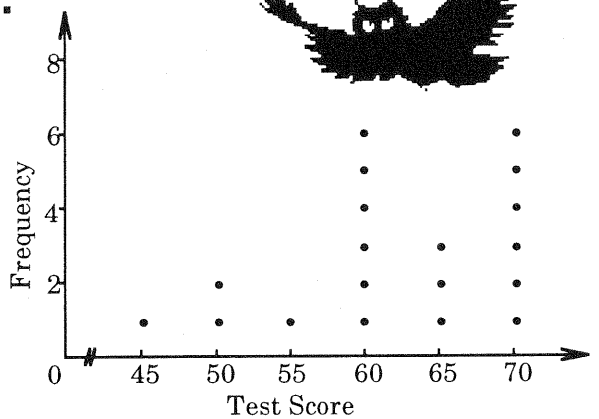
CODE KEY	
57	– B
12	– I
6	– M
38	– A
86	– C
48·5	– E
41·5	– N
37	– O
60	– T
18	– Y
55	– R

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HOW DOES A VAMPIRE BEGIN ITS LETTERS?

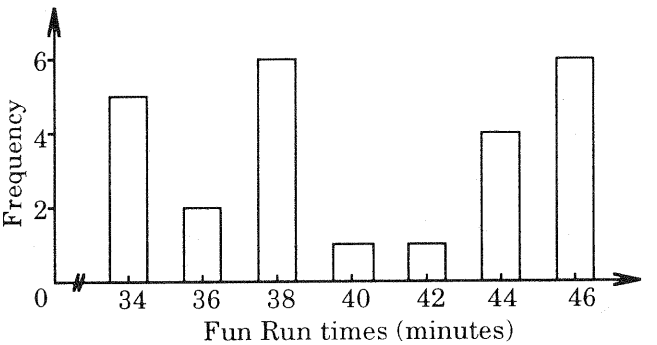


6.

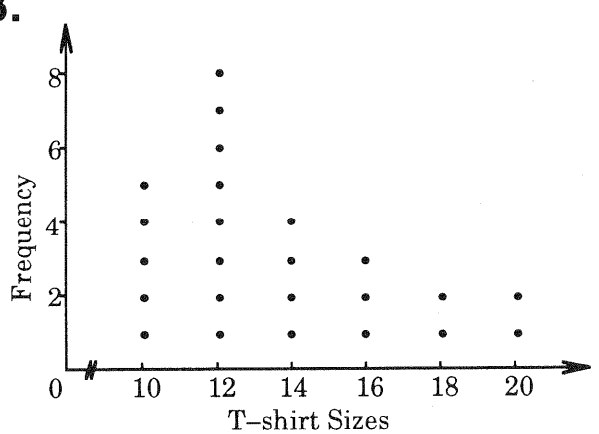




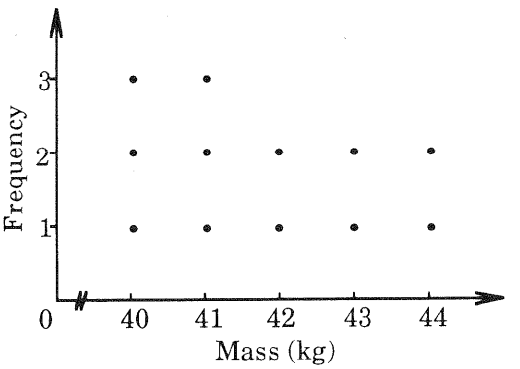
7.



8.



9.



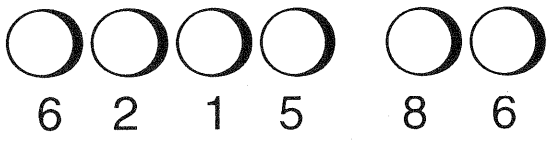
10.

Height (mm)	Frequency
45	2
50	3
55	8
60	2



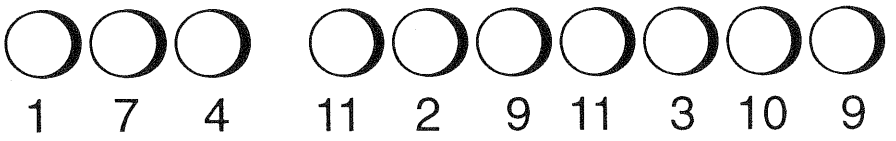
11.

Test scores	Frequency
82	6
84	1
86	3
88	4
90	2



6 2 1 5 8 6

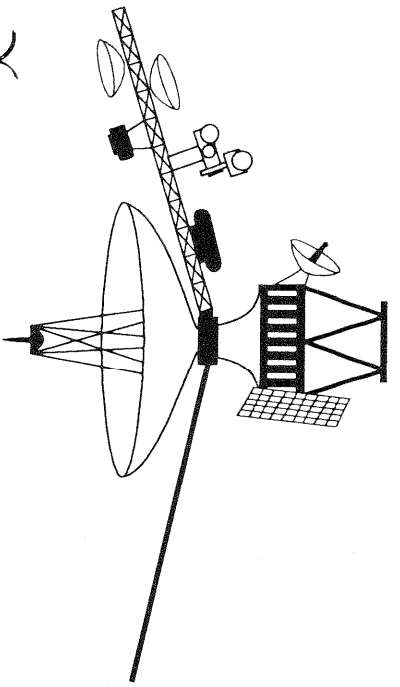




1 7 4 11 2 9 11 3 10 9

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WHY DID THE ASTRONOMER BANG HER HEAD ON THE BRICK WALL IN THE MORNING?

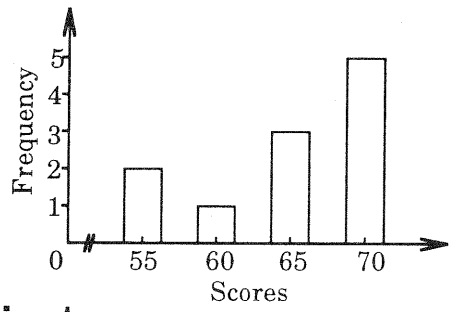


1	2	3	4	5
6	7	8	9	

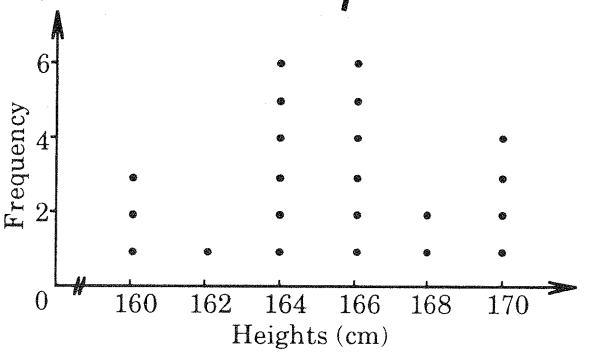
Determine the modes of the following. Look for your answer in the Code Key. Notice the word next to it. Write that word in the numbered puzzle box above.

1. 3, 4, 4, 4, 5, 6, 6, 7
2. 12, 15, 17, 17, 18, 19, 20
3. 19, 31, 18, 30, 18, 30, 17, 18, 31, 30
4. 65, 78, 68, 75, 69, 77, 75, 79, 75, 75

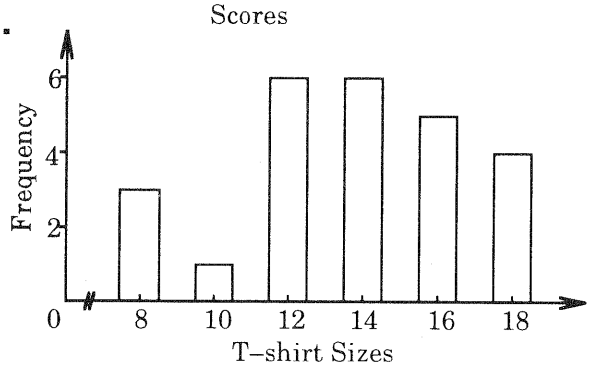
5.



6.



7.



8.

Test Scores	Frequency
55	3
60	14
65	7
70	1
75	9

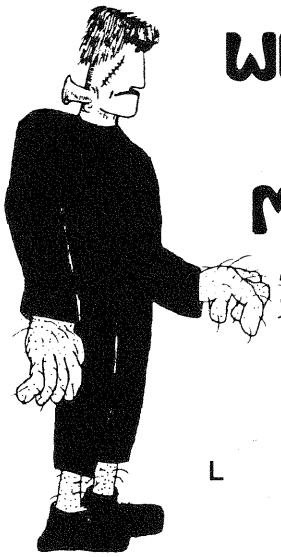
9.

Mass (g)	Frequency
550	6
600	10
650	2
700	1
750	4
800	8

CODE KEY

12, 14 – DURING
17 – SHE
18 – WHEN
600 – DAY
550 – TELESCOPE
75 – TO
60 – THE
70 – SEE
4 – BECAUSE
164, 166 – STARS
18, 30 – WANTED
170 – SKY

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WHAT DID FRANKENSTEIN CALL HIS TEAM OF MONSTER FOOTBALLERS?

Determine the mean, median or mode in each of the following situations. Look for your answers in the code below. Write the letter of the question above its "average".

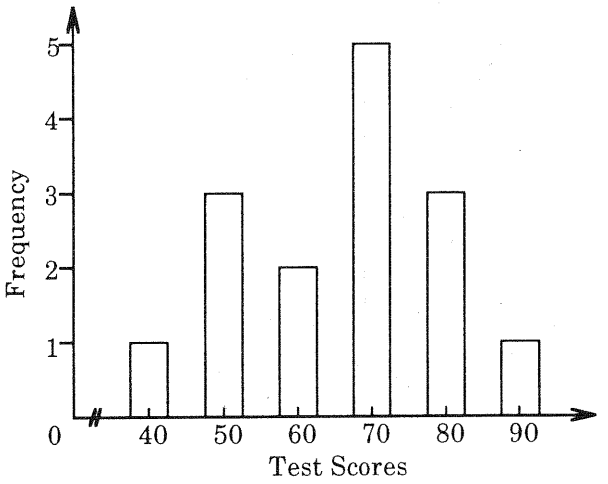
THE _

70 $38 24 32·6 median 66 28 mode

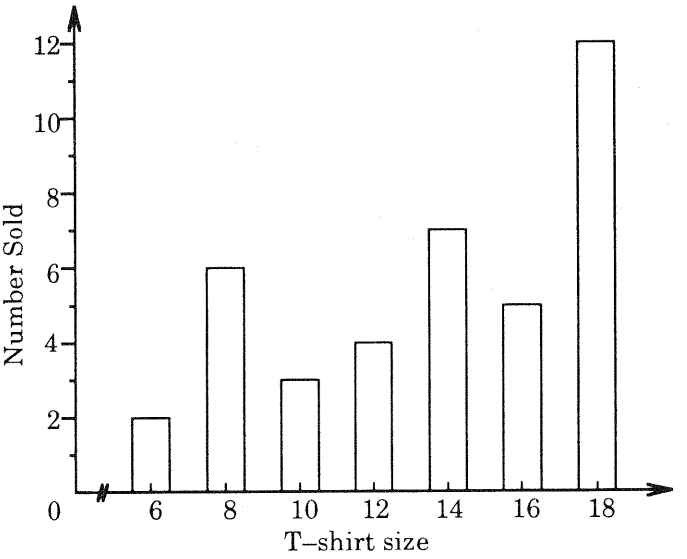
L Five friends going to the Royal Show take spending money of $25, $20, $35, $60 and $50 each. If they decided to pool their money and share the total evenly, how much would each person get?

A The results of a class mental test are shown here. What was the average test score?

A The mode and median mental scores from the graph are the same. What score is that?



S Robyn operates a T-shirt making business from home. In one week her sales are shown. Which "average" would be of most use to her in planning what sizes to make?



S In Perth, over a week, the maximum temperatures were 32·2°, 28·1°, 27·5°, 34·5°, 38·2°, 37·3° and 30·4°. What was the mean maximum temperature for that week?

R In the early games of the cricket season, a cricketer made scores of 20, 33, 12, 14, 57 and 32 in each of 6 innings. What was his mean run score in each innings?

L If the cricketer above made a duck in his next turn at bat, what would the mean score be then?

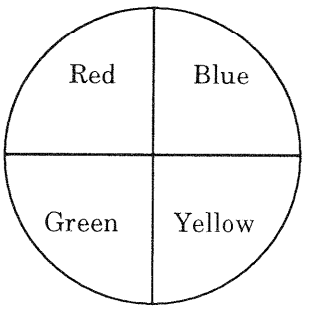
C For the scores 58, 47, 36, 41 and 58, which of the mean, median or mode does 47 represent?

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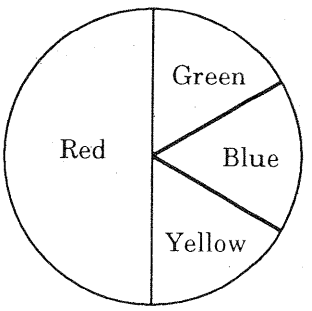
WHAT DO FAMOUS VAMPIRES RECEIVE ON THEIR BIRTHDAYS?

Shown below are some spinners and some frequency tables. Determine the frequency table that might occur if each spinner was spun 100 times. Write the letter of each table above the number of each spinner in the code on the next page.

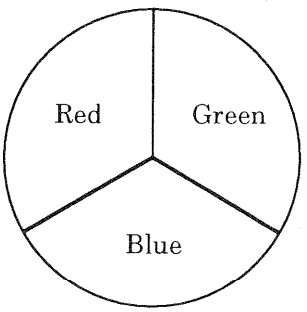
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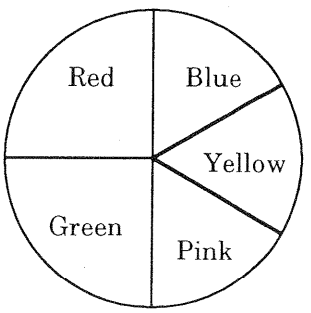
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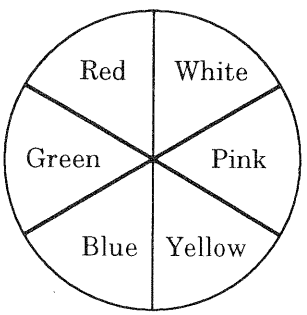
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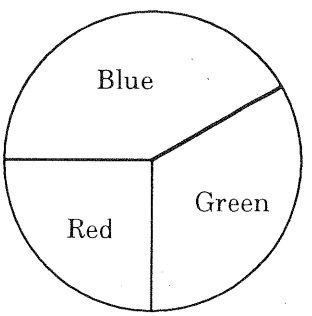
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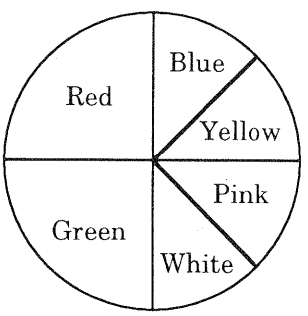
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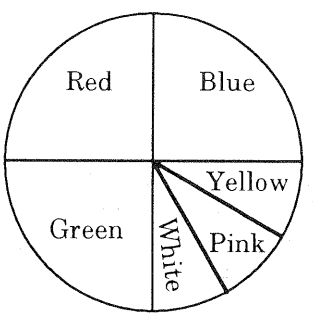
⑥



⑦



⑧



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WHAT DO FAMOUS VAMPIRES RECEIVE ON THEIR BIRTHDAYS?

G

Colour	Frequency
Red	15
Green	18
Blue	16
Yellow	17
Pink	19
White	15

F

Colour	Frequency
Red	24
Green	27
Blue	16
Yellow	15
Pink	18

A

Colour	Frequency
Red	22
Green	26
Blue	25
Yellow	8
Pink	10
White	9

N

Colour	Frequency
Red	30
Green	34
Blue	36

I

Colour	Frequency
Red	53
Green	16
Blue	14
Yellow	17

A

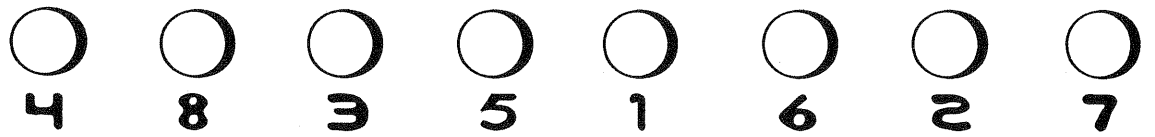
Colour	Frequency
Red	27
Green	35
Blue	38

L

Colour	Frequency
Red	23
Green	25
Blue	13
Yellow	15
Pink	14
White	10

M

Colour	Frequency
Red	24
Green	27
Blue	25
Yellow	24



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MATH O MAGIC 2·3 SOLUTIONS

1. A BITE IN SHINING ARMOUR
2. THE PASSENGER LIST
3. SHE WENT DATA WAY
4. WARTS AND CROSSES
5. NOBODY KNOWS
6. PLEASE BE SHEETED
7. A VAMPIRE THAT SUCKS YOU THROUGH A STRAW
8. ITS DEADY BEAR
9. POPULAR MECHANICS
10. HAUNT AND SEEK
11. HE WENT AS TWINS
12. A DEEP SHE FISH
13. THE FIRST TIME HE SALUTED HE NEARLY KILLED HIMSELF
14. FROM FLYING SAUCERS
15. BY PUBLIC TRANCEPORT
16. A BALLOON WITH THE SKIN PEELED OFF
17. FIRST THEY FRAME ME THEN THEY HANG ME
18. SHE WAS GROPING FOR WORDS
19. HAILING TAXIS
20. SHE BATS HER EYELIDS
21. SO HE COULD BREAK OUT
22. UNDERTIME
23. GHOULASH
24. AN ANIMAL THAT CAN MILK ITSELF
25. AN ANIMAL THAT WAKES PEOPLE WHO LIVE ON THE TOP FLOOR
26. A BIRD THAT CAN ASK ITS WAY HOME IF IT GETS LOST . . .
27. AT A CROW BAR
28. A CAN OPENER
29. THE WEEDER'S DIGEST
30. HE WAS DISCHARGED
31. TOMB IT MAY CONCERN
32. BECAUSE SHE WANTED TO SEE STARS DURING THE DAY
33. THE ALL SCARS
34. FANGMAIL

